Wage Inequality Wars - Part II

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Today

- End of 1990's: consensus on importance of skill biased technological change (SBTC).

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  Card and Di Nardo, 2002

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- Mid 2000's: open war! East-West coast divide

- Key arguments for the critique: institutions, within group inequality, gender wage inequality, racial wage inequality.

East vs. West Coast Battles, Labor Economics style



Card and Di Nardo 2002

- Original title "10 Things we hate about SBTC"
- SBTC provides an incomplete and sometimes undisciplined ("name the residual") account of changes in wage structure.
- Highlight: Problems (patterns in the data inconsistent with SBTC) Puzzles (patterns that could be explained by SBTC but seemed to be due to other factors)

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- Highlight: Problems (patterns in the data inconsistent with SBTC) Puzzles (patterns that could be explained by SBTC but seemed to be due to other factors)
- Main message: economists love uni-causal stories but in real world is really hard to get "uni-causal" implications.

More moments: 1990s era, flat period of inequality?

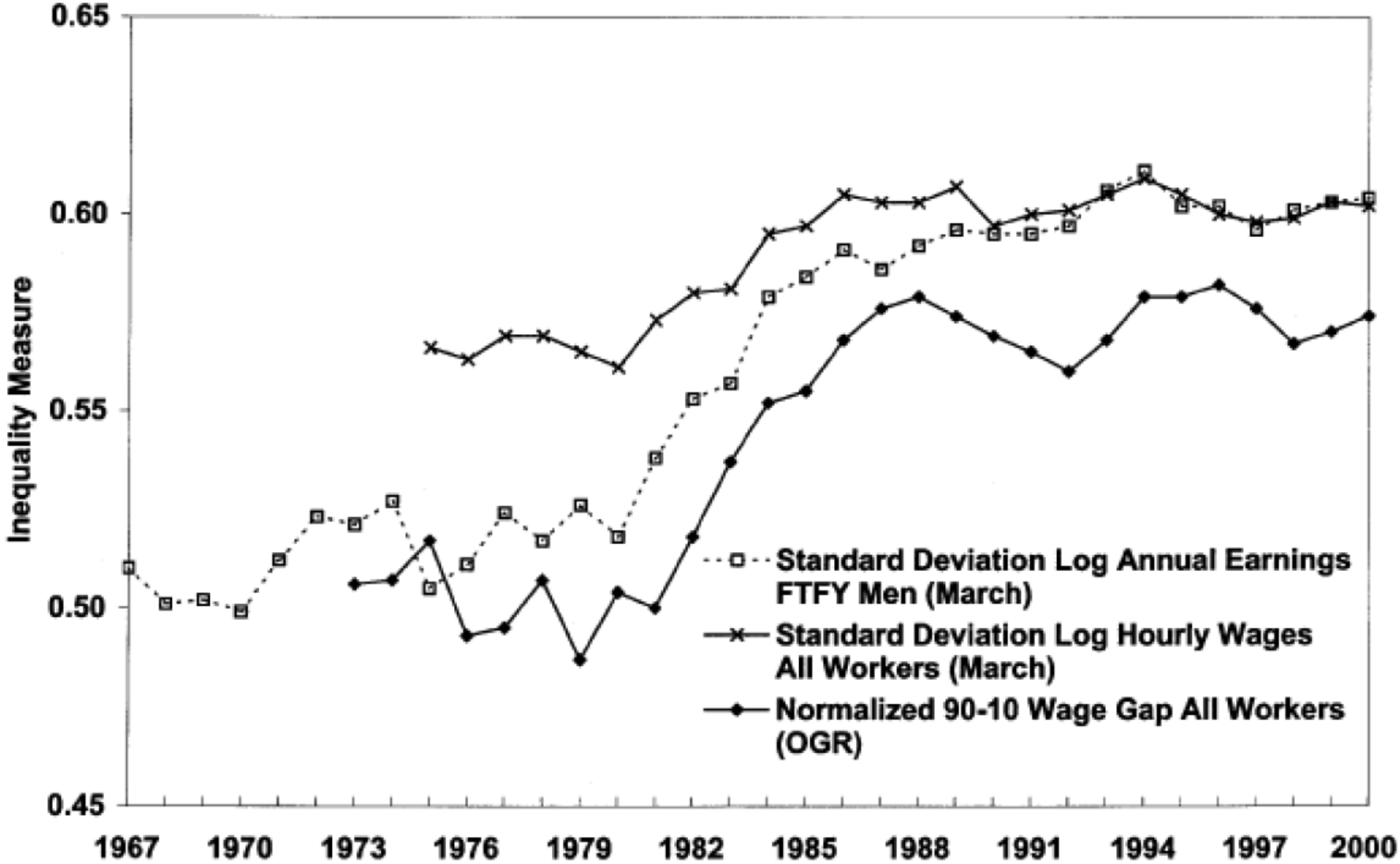


FIG. 2.—Alternative measures of aggregate wage inequality

Were the 1980's an aberration?

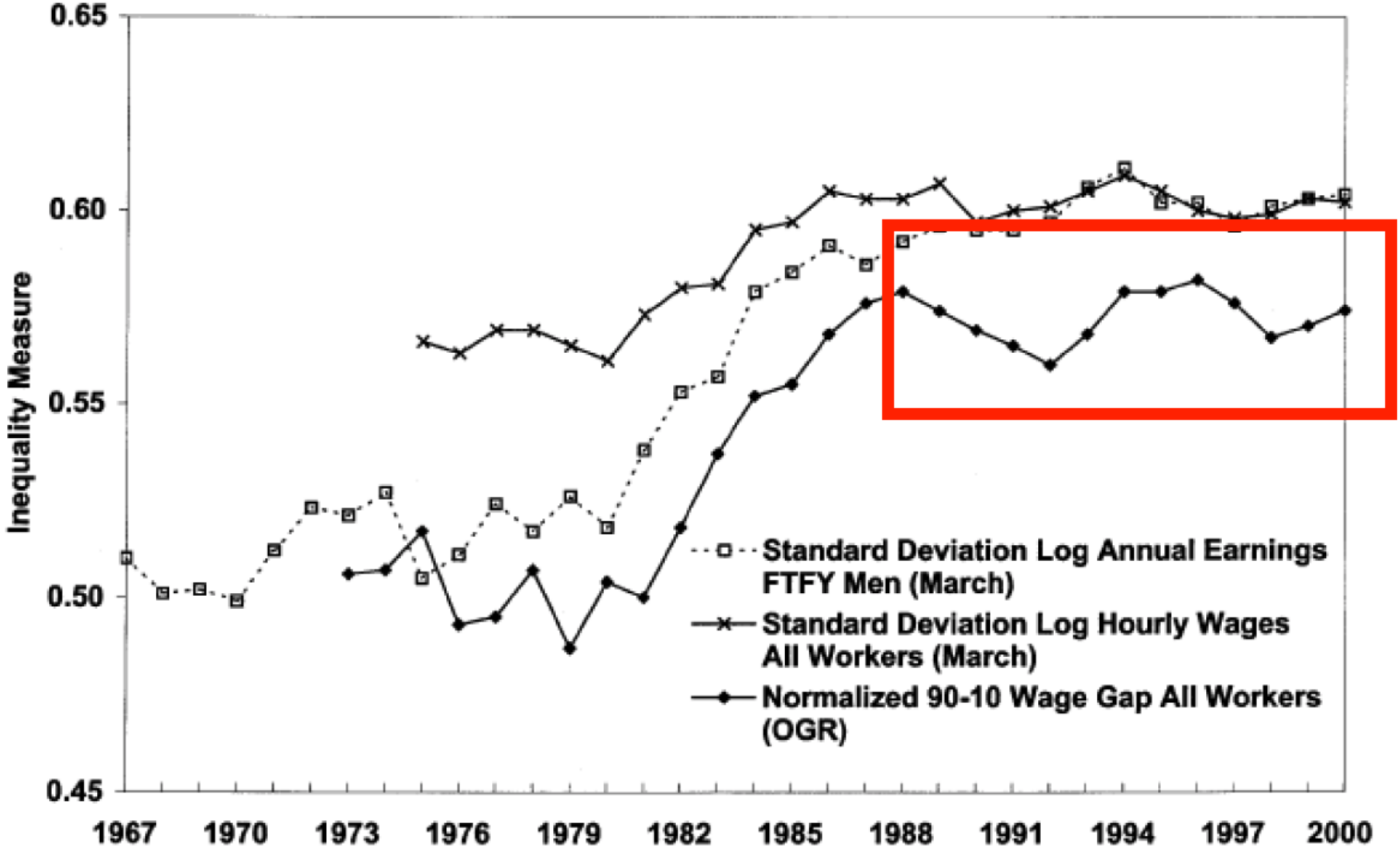


FIG. 2.—Alternative measures of aggregate wage inequality

Technology or Tautology?

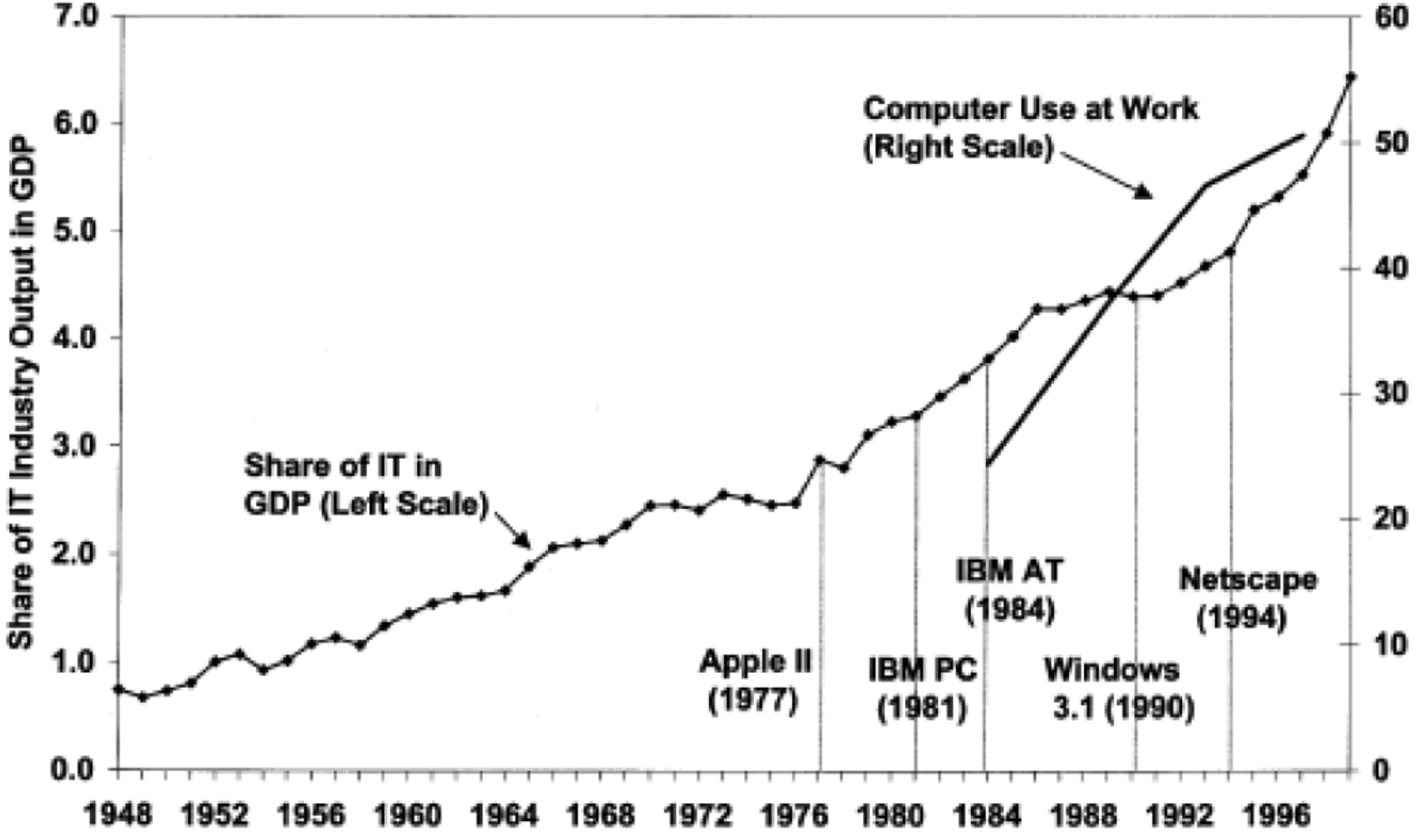


FIG. 1.—Measures of technological change

Several puzzles

Table 2
Summary of Predictions of SBTC for Dimensions of Wage Inequality

Wage Structure Feature	Prediction from Computer-Use-Skill-Complementarity View	Prediction from Rising-Skill-Price View	What Happened?
Overall inequality:			
By decade:			
1970s	?	?	Little change
1980s	Increase	Increase	Large increase
1990s	Increase	Increase	Little change
FTFY men versus all workers, 1980–2000	?	Larger for all	Larger for FTFY
Between group differences:			
By education:			
College–high school, 1980s	Increase	Increase	Large increase
All workers, 1990s	Increase	Increase	Little increase
College–high school by age:			
Younger workers, 1980s onward	Increase	Increase	Increase
Older workers, 1980s onward	Increase	Increase	Little increase
Science versus humanities, 1980s, B.S.-B.A. degree holders	Increase	Increase	Decrease

Conclusions

- 76% of the rise in the 90-10 wage gap b/w 1979-1999 occurred between 1979-1984 $\rightarrow$ first year that the IBM-AT was introduced.
- SBTC may have played a role but it is unlikely to be the only story (too many problems/puzzles associated with that).

Conclusions

- 76% of the rise in the 90-10 wage gap b/w 1979-1999 occurred between 1979-1984 $\rightarrow$ first year that the IBM-AT was introduced.
- SBTC may have played a role but it is unlikely to be the only story (too many problems/puzzles associated with that).
- If it's not SBTC then what is it?

Quite a coincidence!

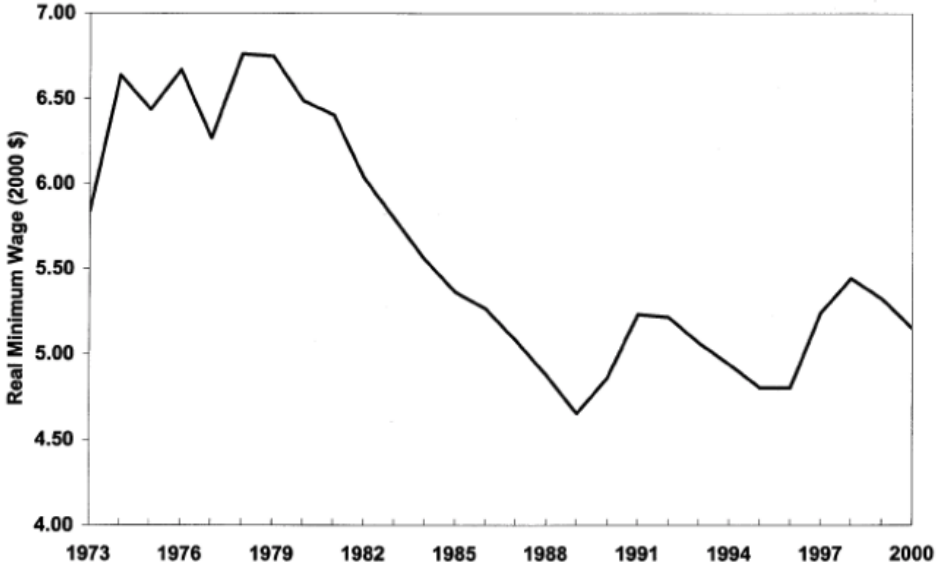


FIG. 22.—Real minimum wage, 1973–2000

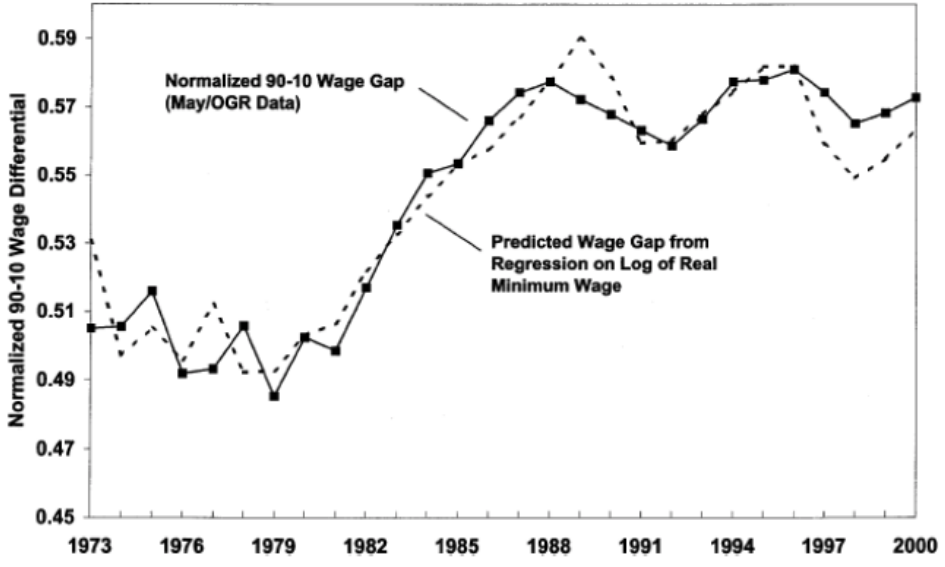


FIG. 23.—Wage inequality and the minimum wage

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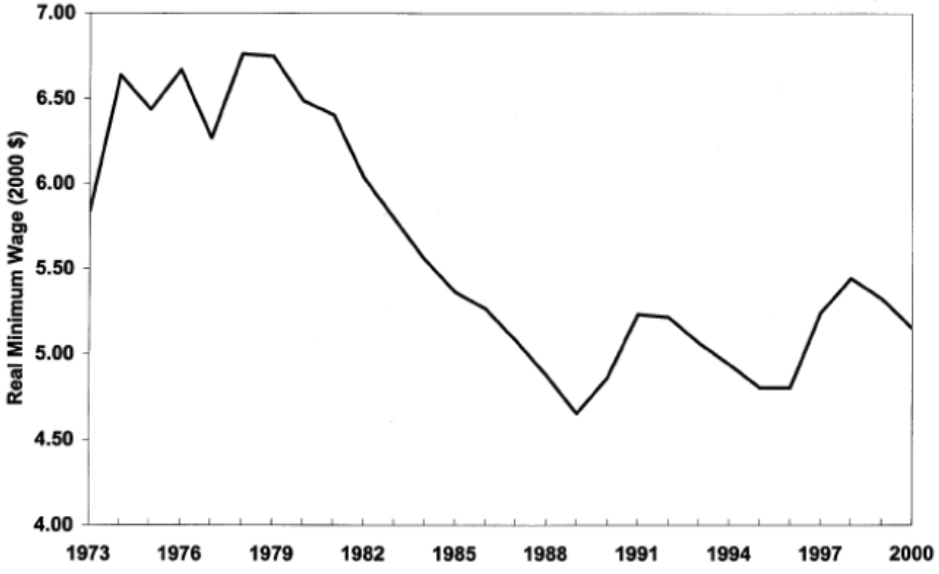


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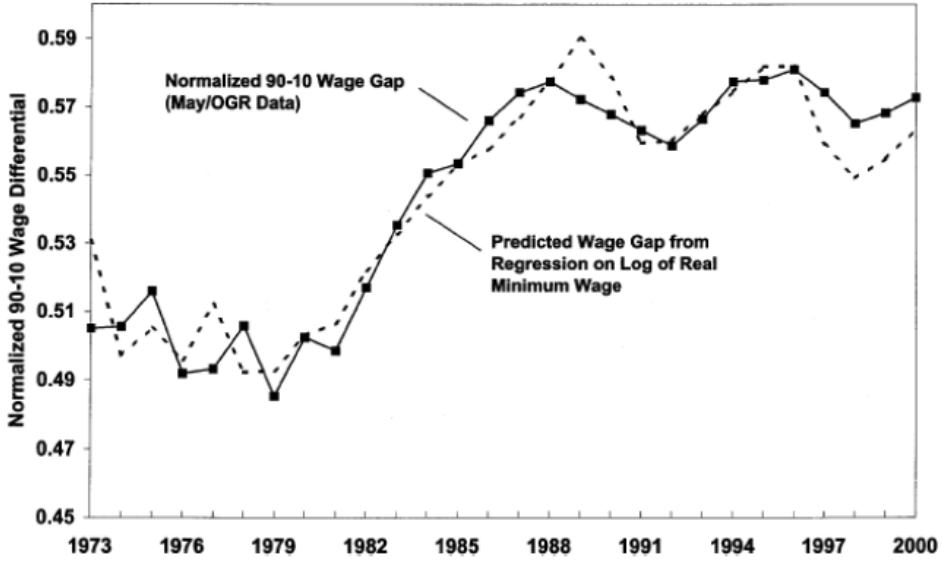
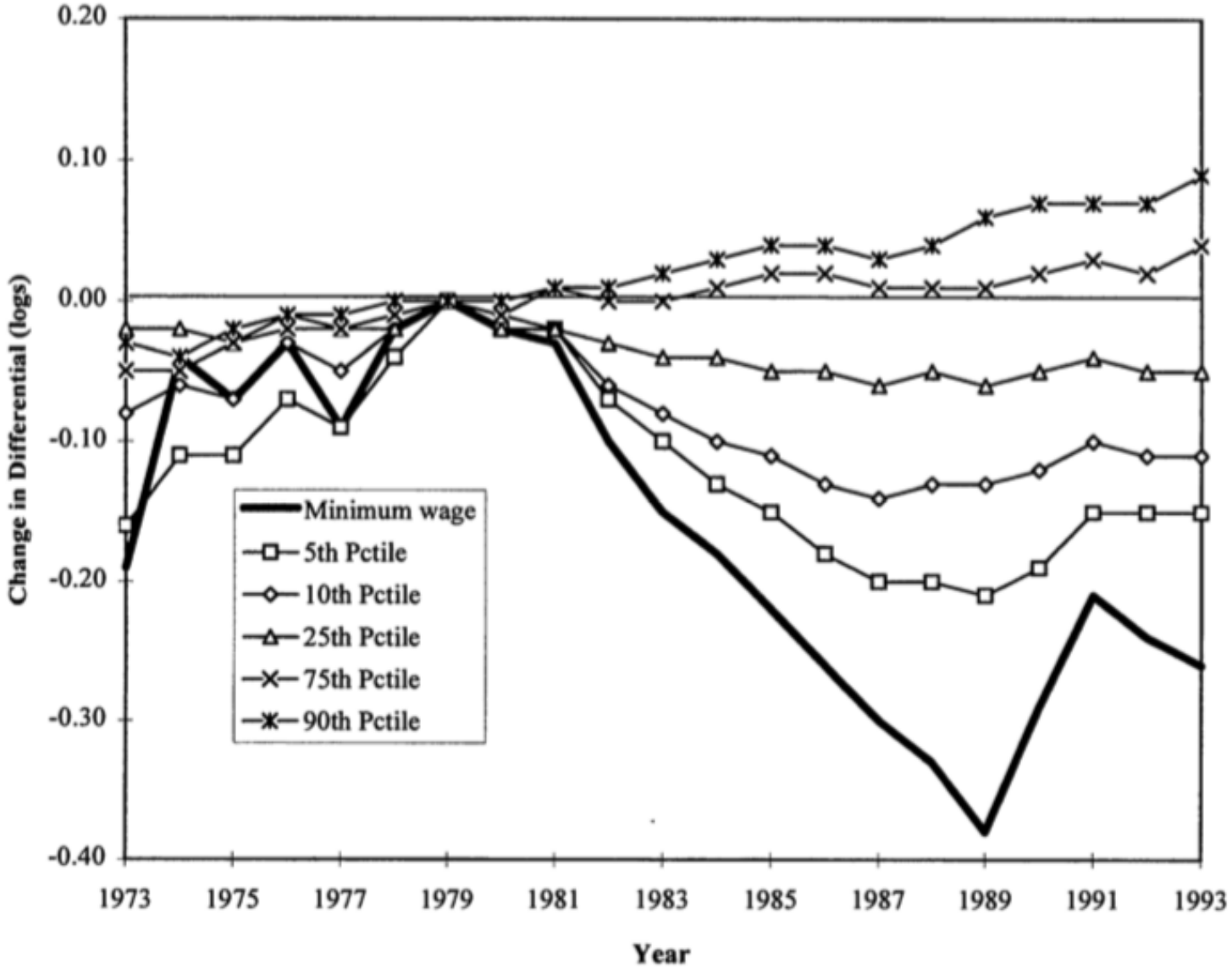


FIG. 23.—Wage inequality and the minimum wage

Lee (1999)


0.20
0.10
0.00
-0.10
-0.20
-0.30
-0.40
Change in Differential (logs)
Minimum wage
5th Pctile
10th Pctile
25th Pctile
75th Pctile
90th Pctile
1973
1975
1977
1979
1981
1983
1985
1987
1989
1991
1993
Year

Card and Lemieux 2001

- Original Title: “It’s supply, stupid”.

Card and Lemieux 2001

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- Starting point: Are we sure we really need technical change to explain the 80s rise in return to schooling?
- What if we are just observing a reduction in relative supply of college grads among recent cohorts?
- In order for the last point to have a "bite" we need to have imperfect substitution across age groups within education category.
- International comparison (USA + UK + Canada).

Starting observation

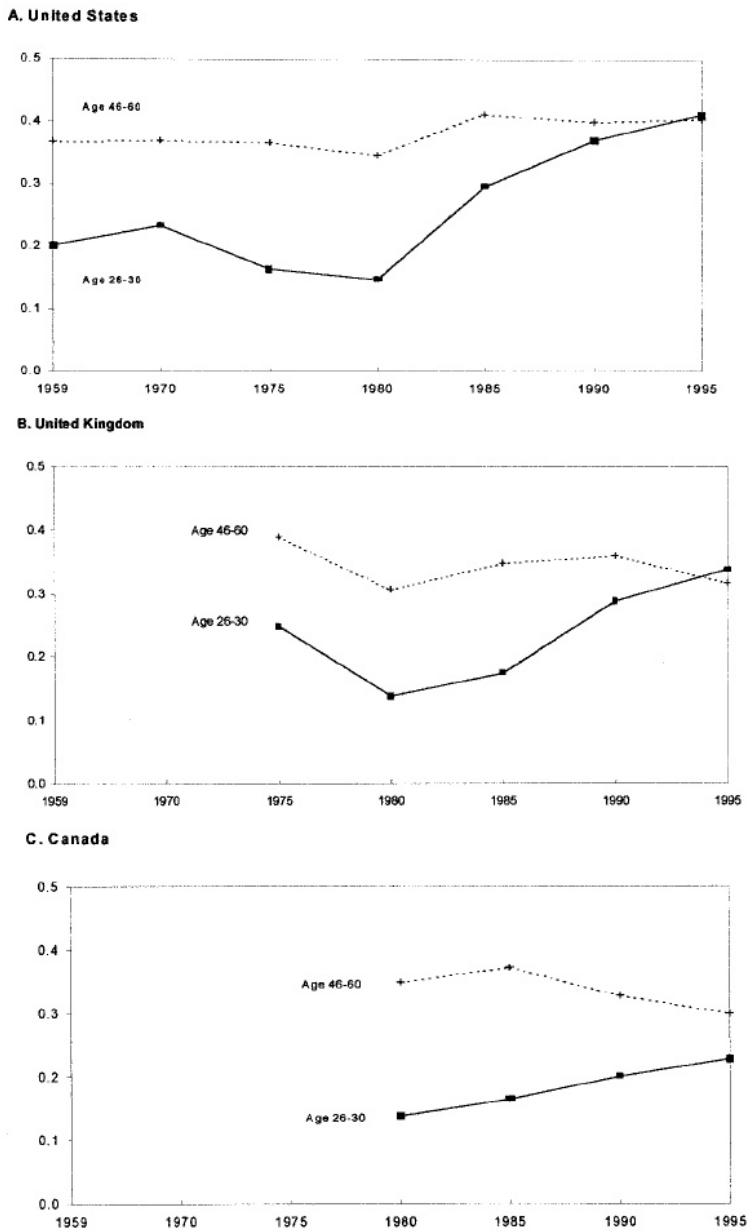


FIGURE I
Estimated College-High School Wage Differentials for Younger and Older Men

Starting observation

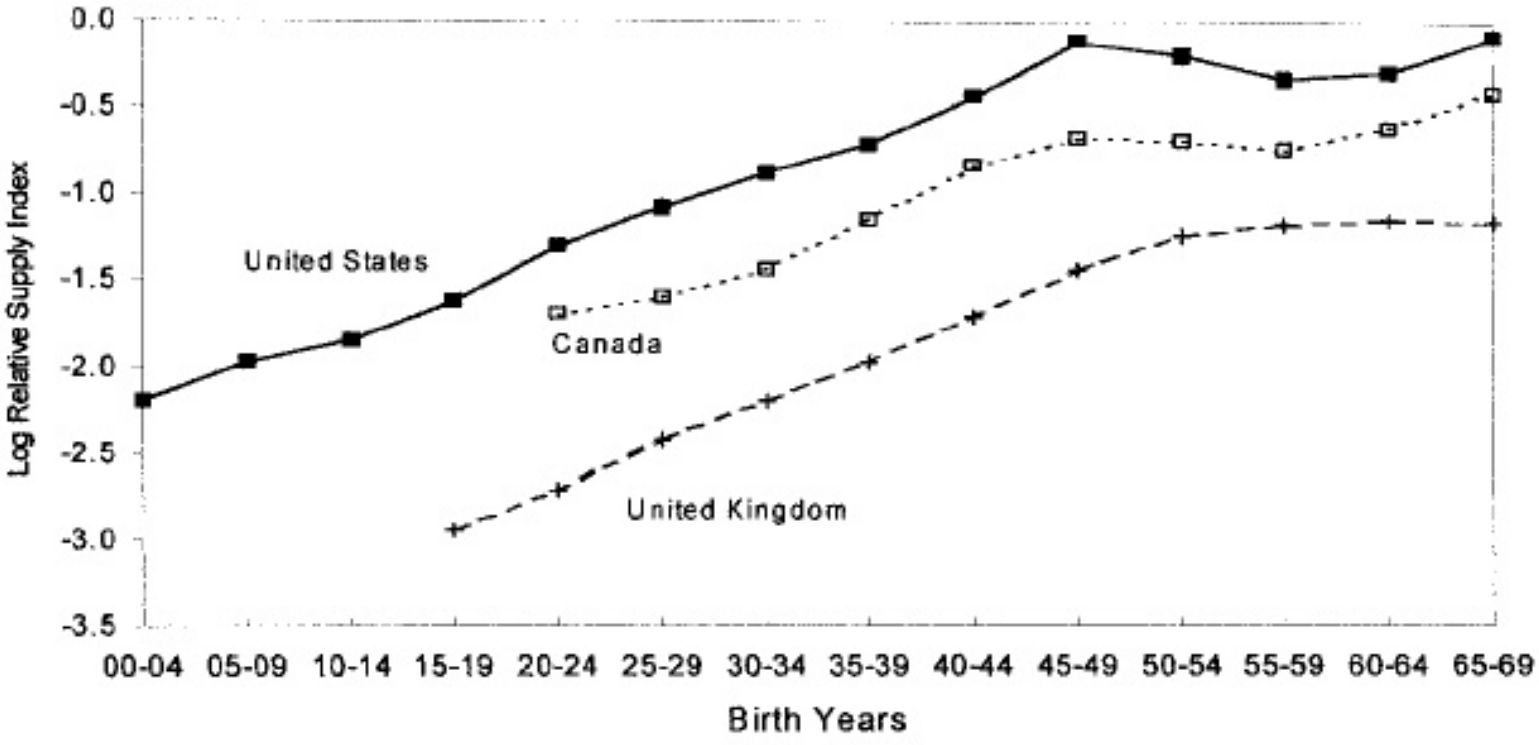


FIGURE IV
Relative Supply of College-Educated Workers by Cohort

Nested CES

- Aggregate production function

$$y_t = \left(\theta_{ht} H_t^{\rho} + \theta_{ct} C_t^{\rho}\right)^{\frac{1}{\rho}} \tag{1}$$

where (H_t, C_t) are high school and college equivalent labor aggregates of age groups j:

$$H_t = \left[\sum_j \left(\alpha_j H_{jt}^{\eta}\right)\right]^{\frac{1}{\eta}} \tag{2}$$

$$C_t = \left[\sum_j \left(\beta_j C_{jt}^{\eta}\right)\right]^{\frac{1}{\eta}} \tag{3}$$

- The relevant elasticities are

$$\sigma_A = (1-\eta)^{-1} \qquad \sigma_E = (1-\rho)^{-1} \tag{4}$$

- The dogma of perfect competition implies that

$$\log \frac{w_{jt}^c}{w_{jt}^h} = \log \frac{\theta_{ct}}{\theta_{ht}} + \log \frac{\beta_j}{\alpha_j} + (\rho - \eta) \log \frac{C_t}{H_t} + (\eta - 1) \log \frac{C_{jt}}{H_{jt}} \tag{5}$$

Interpretation

- It is useful to rearrange the last expression as

$$\log \frac{w_{jt}^c}{w_{jt}^h} = \log \frac{\theta_{ct}}{\theta_{ht}} + \log \frac{\beta_j}{\alpha_j} - \frac{1}{\sigma_E} \log \frac{C_t}{H_t} - \frac{1}{\sigma_A} \left[\log \frac{C_{jt}}{H_{jt}} - \log \frac{C_t}{H_t} \right] \quad (6)$$

  - What if $\eta = 1$?
  - What if $\log \frac{C_{jt}}{H_{jt}} - \log \frac{C_t}{H_t}$ is constant over time?

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  - What if $\eta = 1$?
  - What if $\log \frac{C_{jt}}{H_{jt}} - \log \frac{C_t}{H_t}$ is constant over time?

- How to model relative supplies?
  - Relative supply of college grads for people aged j largely determined by the college enrollment decision that these individuals made when they were 20

$$\log \frac{C_{jt}}{H_{jt}} = \underbrace{\lambda_{t-j}}_{\text{Cohort Effect}} + \underbrace{\phi_j}_{\text{Age}} \quad (7)$$

    age effect is constant over-time!

- So we get that

$$\log \frac{w_{jt}^c}{w_{jt}^h} = \underbrace{\log \frac{\theta_{ct}}{\theta_{ht}}}_{\text{Year Effects}} + \underbrace{\log \frac{\beta_j}{\alpha_j} - \frac{1}{\sigma_A}\phi_j}_{\text{Age}} - \underbrace{(\frac{1}{\sigma_E} - \frac{1}{\sigma_A}) \log \frac{C_t}{H_t}}_{\text{Year Effects}} - \underbrace{\frac{1}{\sigma_A}\lambda_{t-j}}_{\text{Cohort}} \quad (8)$$

Identification

- If there are cohort effects $\rightarrow$ imperfect substitution across age groups $\rightarrow$ Katz and Murphy (1992) is misspecified.

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- If there are cohort effects $\rightarrow$ imperfect substitution across age groups $\rightarrow$ Katz and Murphy (1992) is misspecified.
- But how do you estimate a Nested CES?
- C_t and H_t are model based! Discipline the model to see whether the model derived supply indexes match the data

$$H_t = \left[\sum_j \left(\alpha_j H_{jt}^{\eta} \right) \right]^{\frac{1}{\eta}} \tag{9}$$

$$C_t = \left[\sum_j \left(\beta_j C_{jt}^{\eta} \right) \right]^{\frac{1}{\eta}} \tag{10}$$

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- Look at relative wage across two different age groups *within the same education category* across years

$$\log \frac{w_{jt}^C}{w_{j't}^C} = \log \frac{\beta_j}{\beta_j'} + \psi_t^C - \frac{1}{\sigma_A}\left(\log \frac{C_{jt}}{C_{j't}}\right) \tag{11}$$

$$\log \frac{w_{jt}^H}{w_{j't}^H} = \log \frac{\alpha_j}{\alpha_j'} + \psi_t^H - \frac{1}{\sigma_A}\left(\log \frac{H_{jt}}{H_{j't}}\right) \tag{12}$$

- Using the equation above one can obtain (upon normalization): $(\alpha_j, \beta_j, \sigma_A)$.
- Steps:
  1. Estimate (11) and (12) $\rightarrow (\alpha_j, \beta_j, \sigma_A)$.
  2. Use $(\alpha_j, \beta_j, \sigma_A)$ to construct (H_t, C_t).
  3. Estimate (8).

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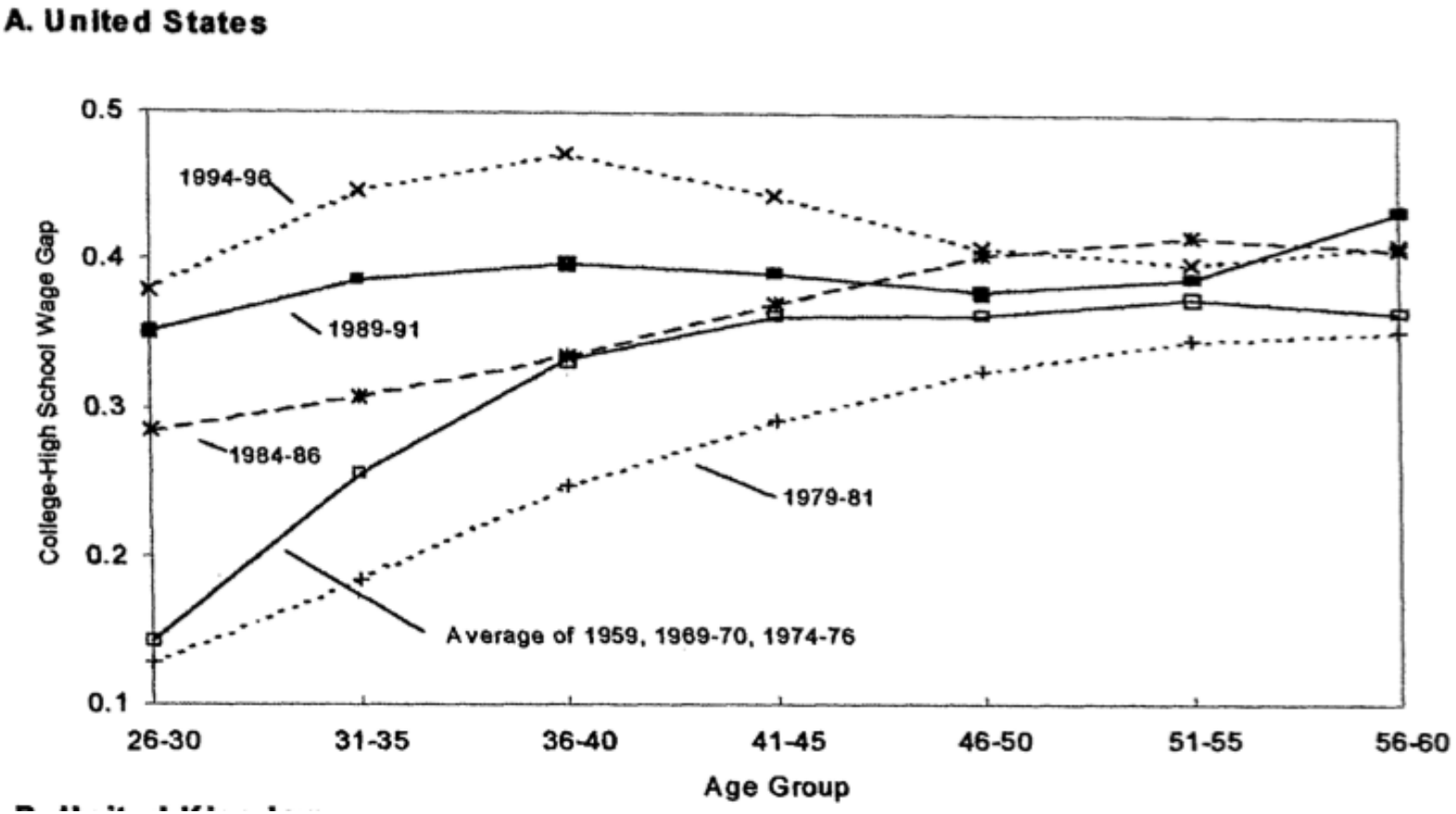
Facts

TABLE I

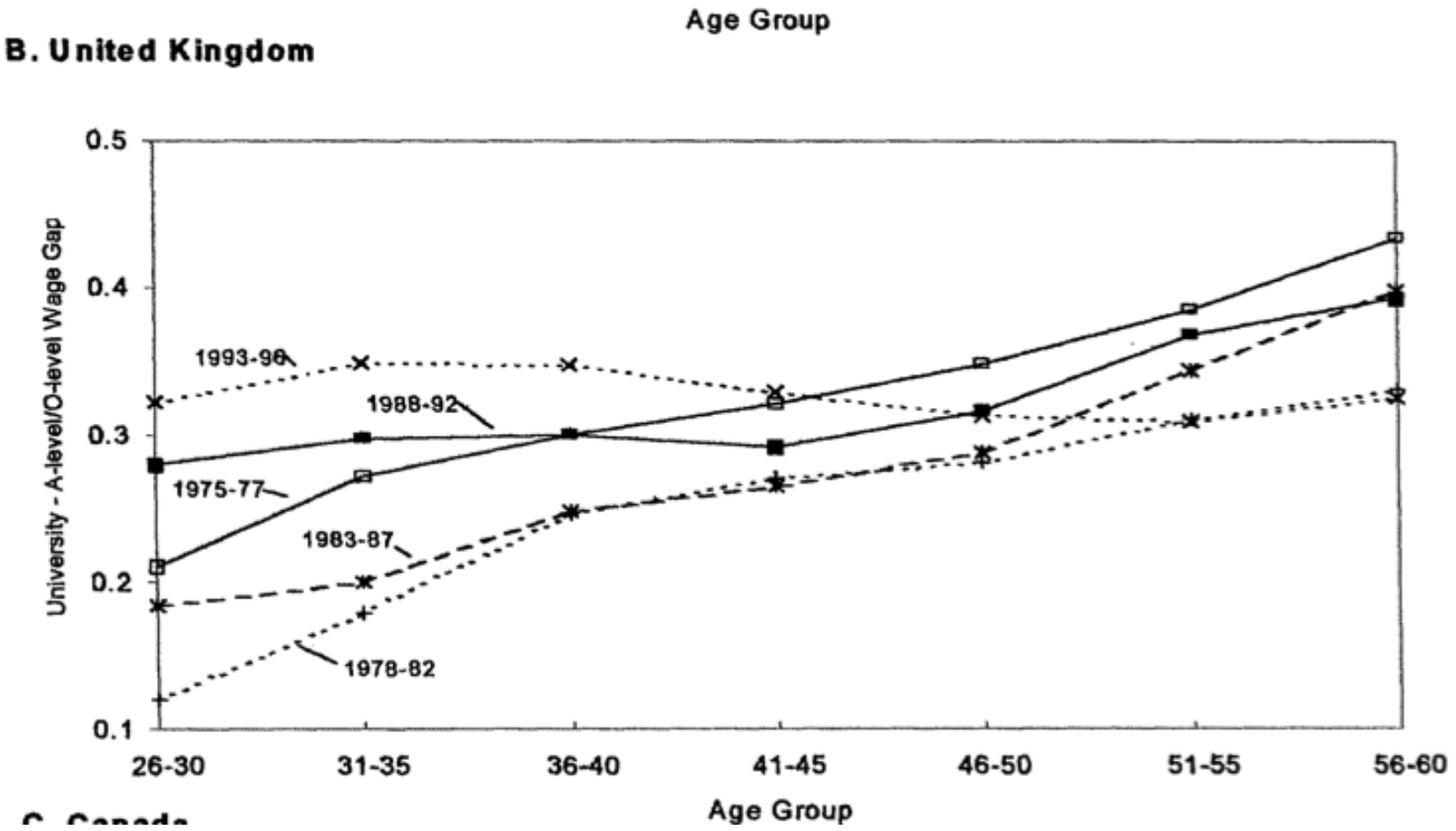
COLLEGE-HIGH SCHOOL WAGE DIFFERENTIALS BY AGE AND YEAR

	Age range						
	26–30	31–35	36–40	41–45	46–50	51–55	56–60
A. United States							
1959	0.136	0.268	0.333	0.349	0.364	0.379	0.362
	(0.007)	(0.007)	(0.008)	(0.011)	(0.013)	(0.016)	(0.021)
1969–1971	0.193	0.272	0.353	0.382	0.360	0.378	0.371
	(0.013)	(0.015)	(0.015)	(0.016)	(0.018)	(0.022)	(0.028)
1974–1976	0.099	0.225	0.310	0.355	0.366	0.369	0.363
	(0.012)	(0.014)	(0.017)	(0.018)	(0.019)	(0.020)	(0.028)
1979–1981	0.111	0.180	0.265	0.281	0.336	0.349	0.355
	(0.011)	(0.012)	(0.015)	(0.017)	(0.017)	(0.018)	(0.021)
1984–1986	0.275	0.315	0.324	0.378	0.402	0.433	0.401
	(0.012)	(0.012)	(0.014)	(0.017)	(0.020)	(0.021)	(0.025)
1989–1991	0.331	0.410	0.392	0.395	0.381	0.357	0.461
	(0.012)	(0.013)	(0.014)	(0.015)	(0.018)	(0.022)	(0.025)
1994–1996	0.346	0.479	0.482	0.443	0.407	0.384	0.421
	(0.014)	(0.014)	(0.015)	(0.017)	(0.017)	(0.023)	(0.030)
B. United Kingdom							
1974–1977	0.172	0.323	0.267	0.338	0.340	0.371	0.455
	(0.026)	(0.034)	(0.046)	(0.049)	(0.057)	(0.059)	(0.086)
1978–1982	0.103	0.173	0.267	0.278	0.259	0.325	0.331

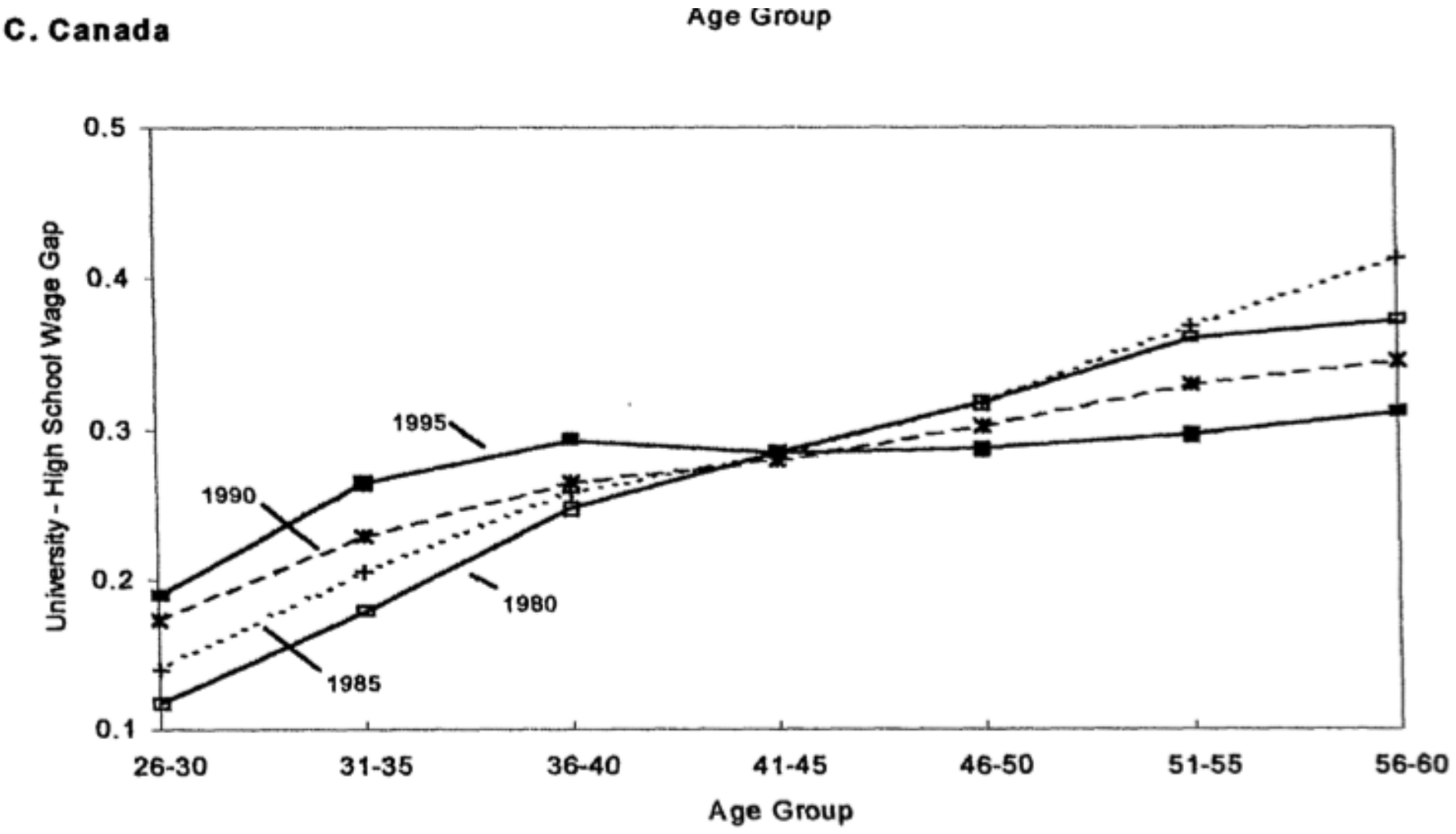
USA


A. United States
College-High School Wage Gap
0.5
0.4
0.3
0.2
0.1
1994-96
1989-91
1984-86
1979-81
Average of 1959, 1969-70, 1974-76
26-30
31-35
36-40
41-45
46-50
51-55
56-60
Age Group

UK


Age Group
B. United Kingdom
0.5
0.4
0.3
0.2
0.1
University - A-level/O-level Wage Gap
1993-96
1988-92
1975-77
1983-87
1978-82
26-30
31-35
36-40
41-45
46-50
51-55
56-60
Age Group

Canada


C. Canada
Age Group
University - High School Wage Gap
0.5
0.4
0.3
0.2
0.1
1995
1990
1985
1980
26-30
31-35
36-40
41-45
46-50
51-55
56-60
Age Group

Cohort effects weaken time trends...

TABLE II

DECOMPOSITIONS OF COLLEGE-HIGH SCHOOL WAGE DIFFERENTIALS BY AGE AND YEAR INTO COHORT, AGE, AND TIME EFFECTS

	United States			United Kingdom			Canada		
	No cohort effects	10 oldest cohorts only	10 oldest coh. eff. same	No cohort effects	7 oldest cohorts only	7 oldest coh. eff. same	No cohort effects	6 oldest cohorts only	6 oldest coh. eff. same
Year effects									
1970	0.026	0.020	0.020	—	—	—	—	—	—
	(0.021)	(0.010)	(0.009)						
1975	−0.020	−0.024	−0.024	0.000	0.000	0.000	—	—	—
	(0.021)	(0.010)	(0.009)						
1980	−0.049	−0.060	−0.062	−0.077	−0.086	−0.076	0.000	0.000	0.000
	(0.019)	(0.011)	(0.009)	(0.026)	(0.021)	(0.018)			
1985	0.058	0.017	0.015	−0.045	−0.057	−0.069	0.020	0.007	−0.004
	(0.020)	(0.013)	(0.010)	(0.027)	(0.025)	(0.021)	(0.019)	(0.017)	(0.013)
1990	0.099	0.022	0.022	0.021	−0.041	−0.037	0.031	−0.011	−0.025
	(0.020)	(0.015)	(0.011)	(0.028)	(0.028)	(0.025)	(0.017)	(0.018)	(0.016)
1995	0.141	0.024	0.034	0.051	−0.060	−0.039	0.043	−0.038	−0.039
	(0.021)	(0.019)	(0.014)	(0.030)	(0.038)	(0.031)	(0.017)	(0.021)	(0.021)
Cohort effects:									
1950–1954	—	—	0.040	—	—	−0.009	—	—	0.028
			(0.011)			(0.019)			(0.015)
1955–1959	—	—	0.124	—	—	0.075	—	—	0.076
			(0.013)			(0.025)			(0.021)
1960–1964	—	—	0.178	—	—	0.134	—	—	0.133
			(0.016)			(0.032)			(0.027)
1964–1969	—	—	0.175	—	—	0.162	—	—	0.142
			(0.024)			(0.046)			(0.036)
Degrees of freedom	36	26	32	24	14	20	18	9	14
χ^2 (p-value)	295.01	48.84	51.09	48.31	10.76	15.33	66.06	12.00	20.51
	(0.00)	(0.00)	(0.02)	(0.00)	(0.70)	(0.76)	(0.00)	(0.21)	(0.11)
R^2	0.87	0.97	0.98	0.77	0.85	0.92	0.89	0.90	0.97

Standard errors are in parentheses. Models are fit by weighted least squares to the age-group by year college-high school wage gaps shown in Table I. Weights are inverse sampling variances of the estimated wage gaps. All models include age effects. For the United States and the United Kingdom, the years indicated when reporting the estimated year effects are the midpoints of the year intervals shown in Table I. The base (omitted) year is 1959 for the United States, 1975–1977 for the United Kingdom, and 1980 for Canada.

All the college premia driven by young guys not a general rise for all age groups!

TABLE II

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Year effects									
1970	0.026	0.020	0.020	—	—	—	—	—	—
	(0.021)	(0.010)	(0.009)						
1975	−0.020	−0.024	−0.024	0.000	0.000	0.000	—	—	—
	(0.021)	(0.010)	(0.009)						
1980	−0.049	−0.060	−0.062	−0.077	−0.086	−0.076	0.000	0.000	0.000
	(0.019)	(0.011)	(0.009)	(0.026)	(0.021)	(0.018)			
1985	0.058	0.017	0.015	−0.045	−0.057	−0.069	0.020	0.007	−0.004
	(0.020)	(0.013)	(0.010)	(0.027)	(0.025)	(0.021)	(0.019)	(0.017)	(0.013)
1990	0.099	0.022	0.022	0.021	−0.041	−0.037	0.031	−0.011	−0.025
	(0.020)	(0.015)	(0.011)	(0.028)	(0.028)	(0.025)	(0.017)	(0.018)	(0.016)
1995	0.141	0.024	0.034	0.051	−0.060	−0.039	0.043	−0.038	−0.039
	(0.021)	(0.019)	(0.014)	(0.030)	(0.038)	(0.031)	(0.017)	(0.021)	(0.021)
Cohort effects:									
1950–1954	—	—	0.040	—	—	−0.009	—	—	0.028
			(0.011)			(0.019)			(0.015)
1955–1959	—	—	0.124	—	—	0.075	—	—	0.076
			(0.013)			(0.025)			(0.021)
1960–1964	—	—	0.178	—	—	0.134	—	—	0.133
			(0.016)			(0.032)			(0.027)
1964–1969	—	—	0.175	—	—	0.162	—	—	0.142
			(0.024)			(0.046)			(0.036)
Degrees of freedom	36	26	32	24	14	20	18	9	14
χ^2 (p-value)	295.01	48.84	51.09	48.31	10.76	15.33	66.06	12.00	20.51
	(0.00)	(0.00)	(0.02)	(0.00)	(0.70)	(0.76)	(0.00)	(0.21)	(0.11)
R^2	0.87	0.97	0.98	0.77	0.85	0.92	0.89	0.90	0.97

Standard errors are in parentheses. Models are fit by weighted least squares to the age-group by year college-high school wage gaps shown in Table I. Weights are inverse sampling variances of the estimated wage gaps. All models include age effects. For the United States and the United Kingdom, the years indicated

Now back-out structural parameters... Begin with CL "first-stage"

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TABLE III
ESTIMATED MODELS FOR THE COLLEGE-HIGH SCHOOL WAGE GAP, BY COHORT AND YEAR

	United States		United Kingdom		Canada	
	(1)	(2)	(3)	(4)	(5)	(6)
Age-group specific relative supply	−0.203	−0.265	−0.233	−0.261	−0.165	−0.161
	(0.019)	(0.026)	(0.058)	(0.071)	(0.042)	(0.040)
Trend	—	0.012	—	0.013	—	0.006
		(0.001)		(0.003)		(0.001)
Year effects:						
1970	0.104	—	—	—	—	—
	(0.012)					
1975	0.124	—	0.0	—	—	—
	(0.017)					
1980	0.129	—	−0.032	—	0.0	—
	(0.019)		(0.023)			
1985	0.255	—	0.060	—	0.029	—
	(0.020)		(0.034)		(0.014)	
1990	0.301	—	0.149	—	0.054	—
	(0.021)		(0.039)		(0.014)	
1995	0.365	—	0.199	—	0.089	—
	(0.023)		(0.044)		(0.017)	
Degrees of freedom	35	40	23	26	17	19
χ^2 (p-value)	66.62	209.34	25.78	52.03	35.00	35.68
	(0.00)	(0.00)	(0.31)	(0.00)	(0.01)	(0.01)
R^2	0.97	0.91	0.86	0.72	0.94	0.94

Standard errors are in parentheses. Models are fit by weighted least squares to the age-group by year college-high school wage gaps shown in Table I. Weights are inverse sampling variances of the estimated wage gaps. All models include age effects. For the United States and the United Kingdom, the years indicated when reporting the estimated year effects are the midpoints of the year intervals shown in Table I.

Closing the circle and estimating equation (8)

TABLE IV

MODELS FOR THE COLLEGE-HIGH SCHOOL WAGE GAP, BY COHORT AND YEAR

	United States			United Kingdom			Canada		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Age-group specific relative supply	−0.202	−0.209	−0.208	−0.233	−0.233	−0.233	−0.166	−0.165	−0.165
	(0.026)	(0.025)	(0.019)	(0.078)	(0.078)	(0.059)	(0.041)	(0.041)	(0.042)
Trend	0.017	0.020	0.015	0.021	0.018	0.020	−0.001	−0.002	−0.006
	(0.002)	(0.002)	(0.002)	(0.007)	(0.006)	(0.005)	(0.007)	(0.015)	(0.024)
1980 dummy	—	—	−0.057	—	—	−0.073	—	—	−0.006
			(0.011)			(0.016)			(0.026)
Katz-Murphy aggr. supply index	−0.414	—	—	−0.466	—	—	0.069	—	—
	(0.047)			(0.156)			(0.247)		
Aggr. supply index for men with imperfect substitution across age groups	—	−0.483	−0.327	—	−0.340	−0.416	—	0.134	0.275
		(0.053)	(0.051)		(0.114)	(0.087)		(0.547)	(0.826)
Degrees of freedom	39	39	38	25	25	24	18	18	17
χ^2 (p-value)	143.05	138.02	81.01	50.39	50.47	27.34	35.08	35.12	35.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.29)	(0.01)	(0.01)	(0.01)
R^2	0.94	0.95	0.96	0.73	0.73	0.86	0.94	0.94	0.94

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No SBTC in Canada?

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Conclusions

- Age groups imperfect substitutes $\sigma_A \in [4; 6]$.
- Back to magical number $\sigma_E \in [1.1; 1.6]$.
- R^2 for the US is 0.95.
- Relative supply explains more of growth in inequality than previously suggested.
- Demand / SBTC somewhat less important than it appears.

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- R^2 for the US is 0.95.
- Relative supply explains more of growth in inequality than previously suggested.
- Demand / SBTC somewhat less important than it appears.
- But why has the supply of college graduates stop increasing?

Discussion

- What explains the slowdown of college graduation rates?
- CL argued that cohorts at the peak of the baby boom (1950s cohorts essentially) were crowded out of college.
- Cohort crowding is present in all three countries.
- In a separate paper CL shows that size of the cohort is able to explain 1/4 of the intercohort slowdown in educational attainment.
- Other theories?

The Harvard Crimson



By Camille G. Caldera

By Jonah S. Berger, Contributing Opinion Writer

Jonah S. Berger '21, a former Associate News editor, is an Economics concentrator in Cabot House. His column appears on alternate Fridays.

February 11, 2021

For decades, the size of Harvard's student body hasn't budged. Despite more and more proof of the sky-high value of a Harvard College degree and the University's continuously growing applicant pool, administrators haven't taken the logical next step: to let in more students.

Harvard now sits fourth among the eight Ivy League schools in its undergraduate enrollment totals, trailing Penn and Cornell substantially, as well as Brown by a small amount.

Yet Harvard might be the only one in this elite club to have seen such stagnant growth in its enrollment for so long. About 2,200 high schoolers were admitted to Harvard's Class of 1982. That number was 2,147 in 1992, 2,074 in the mid-2000s, and just 1,980 for this year's freshman class. Because the yield rate — the percentage of admits who choose to enroll — has risen steadily since the 1980s, the freshman class has stood virtually unchanged at approximately 1,600 students for more than four decades.

Autor, Levy, Murnane (2003)

- Response to critiques of computer use correlations
- Routinization hypothesis: computers substitute for human in the conduct of routine tasks.
- Computers can also complement workers involved in carrying out problem-solving and complex communication activities ("nonroutine" tasks).
- Use Dictionary of Occupational Titles (DOT) to figure out task content of occupations.
- Assess how this influences wage structure following the decrease in the price of computer capital..

The Machine's Eye

- A task is "routine" if it can be accomplished by machines following explicit programmed rules.
- Some tasks are analytical/interactive vs. manual.
- Examples: Bank-Teller; Taxi Driver; Economist;

Manual/Non-Manual Routine/Non-Routine

TABLE I

PREDICTIONS OF TASK MODEL FOR THE IMPACT OF COMPUTERIZATION ON FOUR CATEGORIES OF WORKPLACE TASKS

	Routine tasks	Nonroutine tasks
	Analytic and interactive tasks	
Examples	• Record-keeping • Calculation • Repetitive customer service (e.g., bank teller)	• Forming/testing hypotheses • Medical diagnosis • Legal writing • Persuading/selling • Managing others
Computer impact	• Substantial substitution	• Strong complementarities
	Manual tasks	
Examples	• Picking or sorting • Repetitive assembly	• Janitorial services • Truck driving
Computer impact	• Substantial substitution	• Limited opportunities for substitution or complementarity

Three fundamental Postulates

1. Computer capital is more substitutable for human labor in carrying out routine tasks than nonroutine tasks.
2. Routine and nonroutine tasks are themselves imperfect substitutes.
3. Greater intensity of routine inputs increases the marginal productivity of nonroutine inputs.

A task-based model

- Total output in the economy is a function of computer capital (C), routine (L_R) and non-routine tasks (L_N):

$$Q = (L_R + C)^{1-\beta} L_N^{\beta}$$

- Capital elastically provided at price ρ (this is the exogenous force of the model).
- Aggregate production follows a Cobb-Douglas production where routine tasks and capital are...
- What is the elasticity of substitution between capital/routine and non-routine tasks? Therefore non-routine tasks are...to routine tasks.

Equilibrium

- Workers inelastically supplies one unit of labor.
- These workers are endowed with heterogeneous productivities in routine and non-routine tasks: $E_i = [r_i, n_i]$.
- Endogenously decide to supply R or NR labor based on comparative advantage.

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- Endogenously decide to supply R or NR labor based on comparative advantage.
- How is the equilibrium pinned down?
  - Perfect substitutability of capital and non-routine task implies that

$$w_R = \rho. \tag{13}$$

  - Let $\eta_i = n_i / r_i$. The marginal worker who is indifferent b/w R and NR jobs has an η_i such that

$$\eta^* = w_R / w_N \tag{14}$$

  - Workers with $\eta_i > \eta^*$ supply NR labor and workers with $\eta_i \leq \eta^*$ supply R labor.

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  - Workers with $\eta_i > \eta^*$ supply NR labor and workers with $\eta_i \leq \eta^*$ supply R labor.
  - Given the above, total supply of R and NR labor is given by $g(\eta^*) = \sum_i r_i \mathbf{1}\{\eta_i \leq \eta^*\}$ and $h(\eta^*) = \sum_i n_i \mathbf{1}\{\eta_i > \eta^*\}$.

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  - Productive efficiency requires

$$w_R = \frac{\partial Q}{\partial L_R} = (1-\beta)\theta^{-\beta} \tag{15}$$

$$w_N = \frac{\partial Q}{\partial L_N} = \beta\theta^{1-\beta} \tag{16}$$

$$\theta = \frac{C + g(\eta^*)}{h(\eta^*)} \tag{17}$$

  - Note that θ is the ratio of routine to non-routine tasks.

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$$\theta = \frac{C + g(\eta^*)}{h(\eta^*)} \tag{17}$$

  - Note that θ is the ratio of routine to non-routine tasks.
  - Five equations in five unknowns $(C, w_R, w_N, \eta^*, \theta)$!

Comparative Statics

- Exogenous force in the model: a *decrease* in ρ (price of computer capital).
- This is going to increase the relative demand for Routine-Tasks since

$$\frac{\partial \log \theta}{\partial \log \rho} = -\frac{1}{\beta} \tag{18}$$

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- The relative wage of non-routine jobs is becoming higher $\rightarrow$ more people now supply non-routine jobs $\rightarrow$ increase in demand for routine tasks will be met by computers, not people!
- Although routine labor input declines, an inflow of computer capital more than compensates, yielding a net increase in the intensity of routine task input in production.

Can we derive now predictions at the industry level?

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- Industry j's output (q_j) is a function of routine (r_j) and non-routine (n_j) tasks:

$$q_j = (r_j)^{1-\beta_j} \, n_j^{\beta_j}$$

- Product price is $p_j\,(q_j) \propto q_j^{-\nu}$
- New key level of heterogeneity: $\beta_j \rightarrow$ industry-specific factor share of nonroutine tasks

Industry-Level Demand

- Industry j takes the wages as given and then chooses quantities to maximize profits:

$$
\begin{aligned}
n_j &= w_N^{-1/\nu} \left(\beta_j (1-\nu)\right)^{1/\nu} \left(\frac{w_N}{\rho} \cdot \frac{1-\beta_j}{\beta_j}\right)^{((1-\beta_j)(1-\nu))/\nu} \\
r_j &= \rho^{-1/\nu} \left((1-\beta_j)(1-\nu)\right)^{1/\nu} \left(\frac{w_N}{\rho} \cdot \frac{1-\beta_j}{\beta_j}\right)^{(\beta_j(\nu-1))/\nu}
\end{aligned}
$$

- Additional predictions from fall in price ρ of routine-replacing capital
  1. For a given price decline, the proportionate increase in demand for routine task input is larger in routine-task-intensive (β_j small) industries.
  2. A decline in the price of computer capital also raises demand for nonroutine task input (R and NR are q-complements). This demand increase is proportionately larger in routine-task intensive industries.

Measurement: the Dictionary of Occupational Titles

- Department of Labor examiners physically visit workplaces and score each job using the Handbook for Analyzing Jobs
- Result: more than 12,000 highly detailed occupations rated on 44 dimensions — training times, physical demands, required aptitudes, temperaments, interests
- Editions: 1939, 1949, 1965, 1977, 1991. ALM use the 4th (1977) and revised 4th (1991)
- ALM pick five of the 44 variables, one for each cell of their 2×2 (routine/non-routine $\times$ cognitive/manual), plus non-routine interactive
- Known limitations: thin coverage of services, imprecise constructs, omitted skills

Five variables, five tasks (examples from the Handbook)

- GED-MATH → non-routine analytic: bottom "adds and subtracts 2-digit numbers"; middle "computes discount, interest, profit and loss"; top "oversees analyses of aerodynamic and thermodynamic systems to determine suitability of design for aircraft and missiles"
- DCP, direction/control/planning → non-routine interactive: "plans and designs private residences, office buildings, factories"; "conducts prosecution in court proceedings"; "commands fishing vessel crew"
- STS, set limits/tolerances/standards → routine cognitive: "operates a billing machine to transcribe from office records"; "measures dimensions of bottle using gauges and micrometers to verify setup conforms to specifications"; "prepares and verifies voter lists"
- FINGDEX, finger dexterity → routine manual: "sews fasteners and decorative trimmings to articles"; "feeds tungsten filament wire coils into machine"; "attaches hands to faces of watches"
- EYEHAND, eye-hand-foot coordination → non-routine manual: "drives bus to transport passengers"; "pilots airplane"; "performs gymnastic feats of skill and balance"

Why percentiles?

- A DCP score of 4 is not "twice" a DCP score of 2 — the DOT scales are neither cardinal nor reliably ordinal across variables
- Fix: replace each raw score by its percentile rank in the 1960 distribution of task input across workers
- Why 1960? It predates the computer era, so the yardstick is fixed before the thing being studied starts

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- Why 1960? It predates the computer era, so the yardstick is fixed before the thing being studied starts
- This makes the units readable. "Non-routine analytic input rose 6.8 centiles between 1970 and 1998" means: the job of the average 1998 worker would have sat 6.8 percentiles higher in the 1960 ranking of jobs by maths content

Aggregation: from one examiner's rating to an industry number

- Chain of steps:
  1. Examiner rates a detailed occupation (12,000 of them) on the five variables
  2. Collapse to ≈450 three-digit Census occupation codes (COC)
  3. Attach that occupation's five percentiles to every worker holding it, in Census IPUMS (1960–90) and CPS MORG (1980–98)
  4. Weight each worker by full-time-equivalent hours (sampling weight $\times$ hours $\times$ weeks)
  5. Average up: over everyone $\rightarrow$ the economy-wide series; over workers within an industry $\rightarrow$ 140 consistent industries
- So an industry's task content is just the employment-weighted average task content of the occupations it hires

A concrete example: banking

- 1960: a branch is mostly tellers and bookkeeping-machine operators. Those occupations score high on STS ("operates a billing machine to transcribe from office records") $\rightarrow$ banking's routine-cognitive input is high
- 1998: ATMs and back-office computers. The mix shifts toward loan officers, analysts and managers, who score high on DCP and GED-MATH $\rightarrow$ measured routine-cognitive input falls, non-routine analytic and interactive rise

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- Note what did not happen: no single occupation had to be re-rated. The industry's number moved purely because the occupational mix moved
- That is the extensive margin: changes in the occupational distribution of employment, holding task content fixed at DOT 1977. Measurable 1960–1998
- The intensive margin is the other half: the same occupation re-rated between the 1977 and 1991 editions — the job of a "secretary" itself changed. Measurable 1977–1991

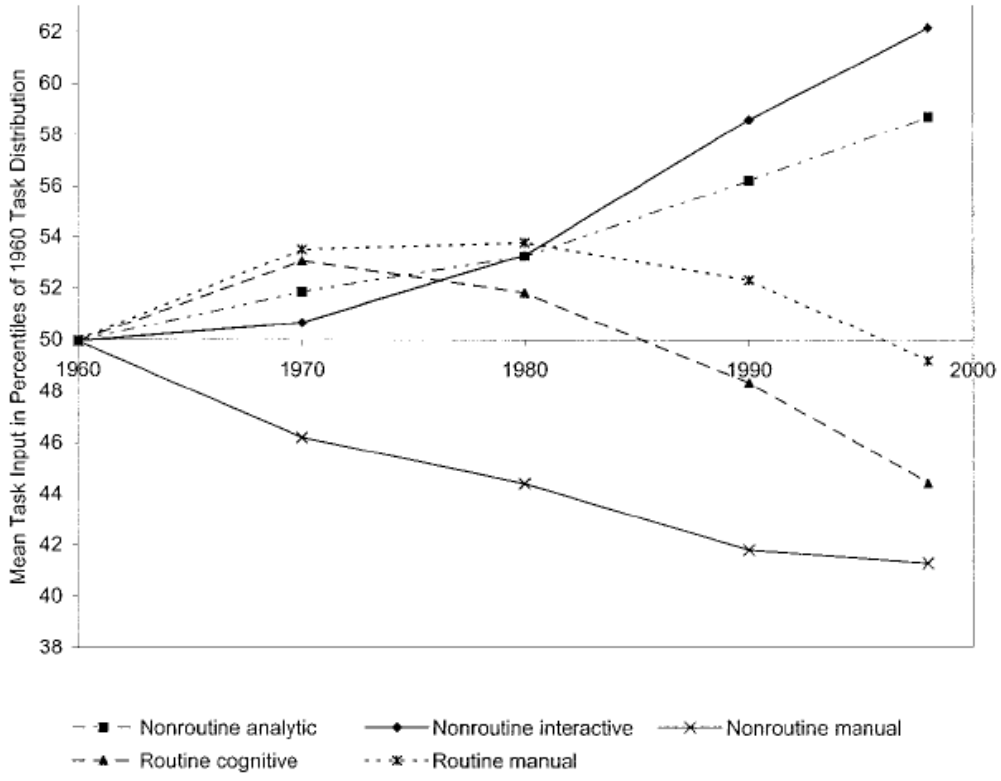


FIGURE I

Trends in Routine and Nonroutine Task Input, 1960 to 1998

Figure I is constructed using *Dictionary of Occupational Titles* [1977] task measures by gender and occupation paired to employment data for 1960 and 1970 Census and 1980, 1990, and 1998 Current Population Survey (CPS) samples. Data are aggregated to 1120 industry-gender-education cells by year, and each cell is assigned a value corresponding to its rank in the 1960 distribution of task input (calculated across the 1120, 1960 task cells). Plotted values depict the employment-weighted mean of each assigned percentile in the indicated year. See Table I and Appendix 1 for definitions and examples of task variables.

Tasks seem to change: is it mainly a between vs. a within industry phenomenon?

TABLE II

MEANS OF TASK INPUT BY DECADE AND DECOMPOSITION INTO WITHIN AND BETWEEN INDUSTRY COMPONENTS, 1960–1998

A. Weighted means of economywide task input by decade (percentiles of 1960 task distribution)

	1. Nonroutine analytic			2. Nonroutine interactive			3. Routine cognitive			4. Routine manual			5. Nonroutine manual		
	All	Male	Fem	All	Male	Fem	All	Male	Fem	All	Male	Fem	All	Male	Fem
1960 (Census)	50.0	54.6	37.0	50.0	58.4	26.4	50.0	48.6	53.8	50.0	43.2	69.2	50.0	56.5	31.6
1970 (Census)	52.5	57.5	41.5	51.1	61.7	27.5	52.2	49.8	57.5	54.0	44.9	74.4	46.9	54.1	30.8
1970 (Census)	51.9	56.4	41.8	50.7	60.8	28.1	53.1	50.8	58.2	53.5	44.7	73.3	46.2	53.2	30.7
1980 (Census)	54.9	58.2	49.2	55.4	63.3	41.8	52.9	50.0	58.0	55.1	44.6	73.3	44.0	53.2	28.1
1980 (CPS)	53.2	56.6	47.9	53.3	61.4	40.4	51.8	50.0	54.8	53.8	43.3	70.4	44.4	55.0	27.5
1990 (CPS)	56.2	57.4	54.6	58.6	62.7	53.0	48.3	48.1	48.6	52.3	42.8	65.5	41.8	53.1	26.3
1998 (CPS)	58.7	59.3	58.0	62.2	63.9	59.9	44.4	46.6	41.6	49.2	41.5	59.3	41.3	52.6	26.5

B. Decomposition of task shifts into between and within industry components for combined genders (10 × annual changes in mean task percentile)

	Total	Btwn	Wthn	Total	Btwn	Wthn	Total	Bwtn	Wthn	Total	Btwn	Wthn	Total	Btwn	Wthn
1960–1970	2.57	1.74	0.83	1.15	−0.34	1.49	2.20	1.14	1.06	4.01	2.39	1.62	−3.03	−2.28	−0.74
1970–1980	3.02	1.54	1.48	4.68	0.26	4.42	−0.14	0.33	−0.47	1.63	0.79	0.84	−2.25	−1.00	−1.25
1980–1990	2.97	0.92	2.05	5.31	0.52	4.79	−3.48	−1.42	−2.07	−1.47	−0.16	−1.31	−2.58	−1.27	−1.31
1990–1998	3.12	0.67	2.45	4.48	0.54	3.94	−4.88	−1.31	−3.57	−3.88	−0.38	−3.50	−0.63	−0.31	−0.31

Sources: Dictionary of Occupational Titles [1977], and all employed workers ages 18–64, Census IPUMS 1960, 1970, 1980, CPS MORG 1980, 1990, and 1998. Samples used for decadal changes in panel B are 1960–1970, 1960 and 1970 Census; 1970–1980, 1970 and 1980 Census; 1980–1990, 1980 and 1990 CPS MORG; 1990–1998, 1990 and 1998 CPS MORG. Two Census 1970 samples are used in panels A and B, one coded for consistency with the 1960 Census occupation codes and a second coded for consistency with the 1980 Census occupation codes. Data are aggregated to 1120 industry-gender-education cells by year and each cell is assigned a value corresponding to its rank in the 1960 distribution of task input (calculated across the 1120, 1960 task cells). Panel A contains the employment-weighted mean of each assigned percentile in the indicated year. Panel B presents a decomposition of the aggregate change in task input over the indicated years into within and between industry components for 140 consistent Census Industry Code (CIC) industries (59 in manufacturing, 81 in nonmanufacturing). See Table I and Appendix 1 for definitions and examples of task variables.

Growth in industry computer use associated w/ shift from routine to non-routine tasks

TABLE III

COMPUTERIZATION AND INDUSTRY TASK INPUT, 1960–1998

DEPENDENT VARIABLE: 10 × ANNUAL WITHIN-INDUSTRY CHANGE IN TASK INPUT, MEASURED IN PERCENTILES OF 1960 TASK DISTRIBUTION

		1. 1990–1998	2. 1980–1990	3. 1970–1980	4. 1960–1970
A. Δ Nonroutine analytic	Δ Computer use 1984–1997	12.04	14.02	9.11	7.49
		(4.74)	(4.97)	(4.17)	(5.28)
	Intercept	0.07	−0.66	−0.26	−0.55
		(1.00)	(1.03)	(0.86)	(1.05)
	R^2	0.04	0.05	0.03	0.01
	Weighted mean Δ	2.45	2.05	1.48	0.83
B. Δ Nonroutine interactive	Δ Computer use 1984–1997	14.78	17.21	10.81	7.55
		(5.48)	(6.32)	(5.71)	(6.64)
	Intercept	1.02	1.46	2.35	0.10
		(1.15)	(1.31)	(1.17)	(1.32)
	R^2	0.05	0.05	0.03	0.01
	Weighted mean Δ	3.94	4.79	4.42	1.49
C. Δ Routine cognitive	Δ Computer use 1984–1997	−17.57	−13.94	−11.00	−3.90
		(5.54)	(5.72)	(5.40)	(4.48)
	Intercept	−0.11	0.63	1.63	1.78
		(1.17)	(1.19)	(1.11)	(0.89)
	R^2	0.07	0.04	0.03	0.01
	Weighted mean Δ	−3.57	−2.07	−0.47	1.06
D. Δ Routine manual	Δ Computer use 1984–1997	−24.72	−5.94	−6.56	4.15
		(5.77)	(5.64)	(4.84)	(3.50)
	Intercept	1.38	−0.16	2.09	0.85
		(1.22)	(1.17)	(0.99)	(0.70)
	R^2	0.12	0.01	0.01	0.01
	Weighted mean Δ	−3.50	−1.31	0.84	1.62

Growth in industry computer use associated w/ shift from routine to non-routine tasks

TABLE III
COMPUTERIZATION AND INDUSTRY TASK INPUT, 1960–1998
DEPENDENT VARIABLE: 10 × ANNUAL WITHIN-INDUSTRY CHANGE IN TASK INPUT, MEASURED IN PERCENTILES OF 1960 TASK DISTRIBUTION

		1. 1990–1998	2. 1980–1990	3. 1970–1980	4. 1960–1970
A. Δ Nonroutine analytic	Δ Computer use 1984–1997	12.04	14.02	9.11	7.49
		(4.74)	(4.97)	(4.17)	(5.28)
	Intercept	0.07	−0.66	−0.26	−0.55
		(1.00)	(1.03)	(0.86)	(1.05)
	R^2	0.04	0.05	0.03	0.01
	Weighted mean Δ	2.45	2.05	1.48	0.83
B. Δ Nonroutine interactive	Δ Computer use 1984–1997	14.78	17.21	10.81	7.55
		(5.48)	(6.32)	(5.71)	(6.64)
	Intercept	1.02	1.46	2.35	0.10
		(1.15)	(1.31)	(1.17)	(1.32)
	R^2	0.05	0.05	0.03	0.01
	Weighted mean Δ	3.94	4.79	4.42	1.49
C. Δ Routine cognitive	Δ Computer use 1984–1997	−17.57	−13.94	−11.00	−3.90
		(5.54)	(5.72)	(5.40)	(4.48)
	Intercept	−0.11	0.63	1.63	1.78
		(1.17)	(1.19)	(1.11)	(0.89)
	R^2	0.07	0.04	0.03	0.01
	Weighted mean Δ	−3.57	−2.07	−0.47	1.06
D. Δ Routine manual	Δ Computer use 1984–1997	−24.72	−5.94	−6.56	4.15
		(5.77)	(5.64)	(4.84)	(3.50)
	Intercept	1.38	−0.16	2.09	0.85
		(1.22)	(1.17)	(0.99)	(0.70)
	R^2	0.12	0.01	0.01	0.01
	Weighted mean Δ	−3.50	−1.31	0.84	1.62

Limitation: measure of computer use comes from CPS and is only available for 80's and 90's. Use National Income and Product Accounts (NIPA), which provides detailed data on capital stocks across 42 major industries excluding government.

Capital less important than computers?

TABLE IV

COMPUTER INVESTMENT, CAPITAL INTENSITY, AND TASK INPUT IN THREE-DIGIT INDUSTRIES 1960–1998: STACKED FIRST-DIFFERENCE ESTIMATES

DEPENDENT VARIABLE: 10 × ANNUAL CHANGE IN QUANTILES OF TASK MEASURE, MEASURED IN PERCENTILES OF 1960 TASK DISTRIBUTION

	A. Δ Nonroutine analytic		B. Δ Nonroutine interactive		C. Δ Routine cognitive		D. Δ Routine manual	
	(1)	(2)	(1)	(2)	(1)	(2)	(1)	(2)
Log(*CI*/*L*)	6.65	6.76	11.59	10.03	−8.27	−8.30	−9.11	−8.20
	(4.13)	(3.97)	(3.21)	(3.31)	(3.63)	(3.26)	(2.57)	(2.29)
	[6.36]	[5.90]	[3.97]	[4.50]	[4.74]	[3.76]	[3.27]	[2.86]
Log(*KI*/*L*)	1.22		−3.41		−2.93		−2.42	
	(4.36)		(4.58)		(4.76)		(3.95)	
	[6.36]		[4.45]		[7.31]		[5.87]	
Δ Log(*K*/*L*)		0.24		3.01		−1.32		−3.89
		(2.35)		(2.24)		(2.12)		(1.92)
		[2.36]		[2.19]		[2.18]		[2.38]
1970–1980 dummy	−0.64	−0.54	1.38	2.49	−0.32	−0.84	0.68	−0.80
	(1.08)	(1.29)	(1.50)	(1.62)	(1.31)	(1.58)	(0.96)	(1.17)
	[1.02]	[1.23]	[2.07]	[2.12]	[1.13]	[0.93]	[0.94]	[1.03]
1980–1990 dummy	−0.34	−0.25	0.58	1.83	−1.62	−2.14	−1.32	−2.90
	(1.57)	(1.60)	(1.81)	(1.70)	(1.56)	(1.86)	(1.11)	(1.38)
	[1.43]	[1.67]	[1.58]	[1.46]	[1.33]	[1.35]	[0.86]	[1.07]
1990–1998 dummy	−1.19	−1.13	−1.91	−0.90	−1.33	−1.71	−1.15	−2.36
	(1.55)	(1.62)	(1.83)	(1.70)	(1.66)	(1.64)	(1.32)	(1.47)
	[1.77]	[1.93]	[1.85]	[1.88]	[1.93]	[1.63]	[0.95]	[1.13]
Intercept	8.89	8.23	12.40	11.30	−9.29	−7.09	−9.62	−5.55
	(4.08)	(4.42)	(4.25)	(3.59)	(4.12)	(3.63)	(3.14)	(2.67)
	[5.45]	[6.38]	[4.76]	[4.80]	[4.48]	[4.36]	[3.75]	[3.30]
R^2	0.06	0.06	0.11	0.12	0.14	0.14	0.20	0.21

	Weighted mean of dependent variable			
1960–1970	1.16	1.74	1.30	1.63
1970–1980	1.23	4.59	−0.20	0.98
1980–1990	2.07	4.69	−2.05	−1.74
1990–1998	2.15	3.76	−3.03	−2.82

n = 492. Robust standard errors in parentheses are heteroskedasticity consistent and account for clustering of errors within 42 consistent NIPA sectors in each decade (168 clusters). Standard errors in brackets additionally account for potential serial correlation within sectors (42 clusters). Each column presents a separate OLS regression of ten times annual industry changes in task input on the indicated covariates. Sample is 123 consistent CIC industries, with four observations per industry. 1960–1970 and 1970–1980 changes use Census IPUMS samples, and 1980–1990 and 1990–1998 use CPS MORG samples. Estimates are weighted by mean industry share of total employment (in FTEs) over the endpoints of the years used to form the dependent variable. All capital measures are in millions of real 1996 dollars.

Log(*CI*/*L*) and Log(*KI*/*L*) are, respectively, one-tenth the log of annual computer investment per FTE and total capital investment per FTE between the two end years used to form the dependent variable. Means of Log(*CI*/*L*) are −1.08, −0.95, −0.87, and −0.73 in 1960–1970, 1970–1980, 1980–1990, and 1990–1998, respectively. Means of Log(*KI*/*L*) are −0.57, −0.54, −0.54, and −0.52 in the corresponding years.

Δ Log(*K*/*L*) is ten times the annual change in log capital/FTE over the two end years used to form the

Patterns hold also when looking within Occupations

TABLE VI
COMPUTERIZATION AND CHANGES IN JOB TASK CONTENT WITHIN OCCUPATIONS 1977–1991
DEPENDENT VARIABLE: 10 × ANNUAL WITHIN-OCCUPATION CHANGE IN QUANTILE OF TASK MEASURE, MEASURED IN PERCENTILES OF 1984 TASK DISTRIBUTION

	A. Δ Nonroutine analytic			B. Δ Nonroutine interactive			C. Δ Routine cognitive			D. Δ Routine manual		
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
Δ Computer use	2.94	3.57	4.02	5.70	5.86	7.08	−18.18	−16.56	−18.48	1.74	0.83	0.37
1984–1997	(1.84)	(1.92)	(2.06)	(1.88)	(1.97)	(2.11)	(3.29)	(3.41)	(3.65)	(2.89)	(3.01)	(3.23)
Δ College grad emp.		−4.79	−4.83		−4.47	−4.58		22.59	22.76		−16.07	−16.03
1984–1997		(5.54)	(5.54)		(5.68)	(5.67)		(9.86)	(9.85)		(8.70)	(8.71)
Δ HS grad emp.		2.83	3.09		−0.19	0.52		16.97	15.86		−10.42	−10.70
1984–1997		(3.78)	(3.81)		(3.88)	(3.90)		(6.73)	(6.77)		(5.94)	(5.99)
Δ Female emp.			−2.37			−6.47			10.14			2.47
1984–1997			(3.94)			(4.03)			(6.99)			(6.19)
Intercept	−0.92	−0.91	−0.95	−0.46	−0.42	−0.52	0.56	0.14	0.30	0.42	0.70	0.74
	(0.40)	(0.41)	(0.41)	(0.41)	(0.42)	(0.42)	(0.71)	(0.72)	(0.73)	(0.63)	(0.64)	(0.64)
R^2	0.01	0.01	0.01	0.02	0.02	0.03	0.06	0.08	0.08	0.00	0.01	0.01
Weighted mean Δ		−0.39			0.58			−2.76			0.74	

n is 470 consistent three-digit Census Occupation Code (COC) occupations. Standard errors are in parentheses. Each column presents a separate OLS regression of ten times the annual change in the occupational task measure (measured in centiles of the 1984 distribution) between the 1977 and 1991 DOT revisions. Computer use, college graduate, high school graduate, and female employment shares are measured as ten times the annual change in the relevant measure from the 1984 and 1997 CPS. Weighted means are 0.183, 0.017, −0.015, and 0.017, respectively. Omitted education categories are some college and high school dropout. Estimates are weighted by the average occupational share of U. S. employment in 1984 and 1997. See Table I and Appendix 1 for definitions and examples of task variables.

Does any of this matter?

- But how relevant are these shifts in tasks for a quantity like the college wage premium (a key quantity analyzed since KM 1992).
- Need to provide a mapping between tasks to demand of college graduates.

Does any of this matter?

- But how relevant are these shifts in tasks for a quantity like the college wage premium (a key quantity analyzed since KM 1992).
- Need to provide a mapping between tasks to demand of college graduates.
- Plan: (i) recall what KM's demand index actually is; (ii) notice it has a piece that tasks should move; (iii) measure that piece directly; (iv) see how much of KM's number it accounts for.

First: what is KM's D_t?

- ALM write the aggregate production function with two technology terms

$$Q_t = \left[\alpha_t (a_t N_{ct})^{\rho} + (1-\alpha_t)(b_t N_{ht})^{\rho}\right]^{1/\rho}, \qquad \sigma = \frac{1}{1-\rho}$$

- α_t: how work activities are allocated between college and non-college labor. This is the task margin. When computers take routine tasks away from non-college workers, α_t moves
- Equating wages to marginal products and taking the ratio, the common term cancels:

$$\ln \frac{N_{ct}}{N_{ht}} = \underbrace{\sigma \ln \frac{\alpha_t}{1-\alpha_t} + (\sigma - 1) \ln \frac{a_t}{b_t}}_{D_t} - \sigma \ln \frac{w_{ct}}{w_{ht}}$$

- Rearranged: $D_t = \ln \frac{N_{ct}}{N_{ht}} + \sigma \ln \frac{w_{ct}}{w_{ht}}$. Given observed quantities, observed wages and an assumed σ, this is arithmetic — nothing is estimated
- So D_t is a residual: whatever makes the data sit on the demand curve. This is exactly the "name the residual" that Card and DiNardo complain about

Step 1: what is the mapping between b/w tasks and relative demand for college grads

- Cross-section, at the midpoint of the sample, across j = industries (or occupations):

$$\text{CollegeShare}_j = \alpha + \sum_{k=1}^{4} T_j^k \pi_k + e_j$$

- CollegeShare_j: college-equivalent share of employment (FTEs) in j; T_j^k: task input in centiles
- $\widehat{\pi}_k$ answers: if an industry does more of task k, how much more college labor does it hire?
- Recall that these $\widehat{\pi}_k$ are then held fixed: an industry's college share responds to its tasks but not to wages. Firms never substitute between college and non-college labor when the relative wage moves. That is exactly $\sigma = 0$ — what ALM call the "fixed-coefficient" specification

$$\Delta D = \underbrace{\Delta \ln \frac{N_{ct}}{N_{ht}}}_{\text{quantity}} + \sigma \underbrace{\Delta \ln \frac{w_{ct}}{w_{ht}}}_{\text{price}}$$

- The task calculation $\sum_k \Delta T_j^k \widehat{\pi}_k$ predicts a quantity — a change in the college employment share. Calling that ΔD drops the price term, i.e. sets $\sigma = 0$
- The college premium rose over 1970–98, so the dropped term is positive and ΔD is understated

Step 2: feed the task changes through the price list

$$\Delta \widehat{\text{CollegeShare}}_{j,1970-1998} = \sum_{k=1}^{4} \Delta T^{k}_{j,1970-1998} \, \widehat{\pi}_{k}$$

- Two versions of ΔT^{k}_{j}:
  - observed task change (Table VII panel A)
  - only the part predicted by computer adoption (panel B) — i.e. the fitted values from regressing task change on computerization
- Aggregate up, convert into an annual change in log college/non-college employment, and you have a task-based number in the same units as D_t
- Now the two can be compared. That is the whole exercise

Why whatever we get is a lower bound

- Same point, in the language you already know — omitted variable bias, long vs short. The long relationship is $\Delta D = \beta_T \Delta T + \beta_w \Delta \ln w$ with $\beta_w = \sigma > 0$; ALM run the short one

$$\widetilde{\beta}_T = \beta_T + \underbrace{\sigma}_{\text{coeff. on omitted}} \times \underbrace{\frac{\text{Cov}(\Delta T, \Delta \ln w)}{\text{Var}(\Delta T)}}_{\text{auxiliary regression}}$$

- Computerization raises non-routine task content and bids up the college wage, so the covariance is positive and $\sigma > 0$
- Unusually, the omitted channel pushes the same way as the included one. Dropping it makes ALM's answer too small, not too big

Table VII, panels A–C: the task changes themselves

TABLE VII
SHIFTS IN COLLEGE-EQUIVALENT LABOR DEMAND IMPLIED BY CHANGES IN JOB TASKS, 1970–1998

	1. 1970–1980 extensive margin	2. 1980–1990 extensive margin	3. 1990–1998 extensive margin	4. 1970–1998 extensive margin	5. 1980–1998 extensive margin	6. 1980–1998 intensive margin	7. 1980–1998 extensive + intensive
A. 10 × observed annual changes in DOT task measures (percentile changes relative to 1960 task distribution)							
Nonroutine analytic	3.02	2.97	3.12	3.04	3.05	−0.39	2.67
Nonroutine interactive	4.68	5.31	4.48	4.84	4.85	0.58	5.43
Routine cognitive	−0.14	−3.48	−4.88	−3.03	−4.26	−2.76	−7.02
Routine manual	1.63	−1.47	−3.88	−1.44	−2.81	0.74	−2.07
B. 10 × predicted annual changes in DOT task measures (percentile changes relative to 1960 task distribution)							
	NIPA computer input measure				CPS computer use measure		
Nonroutine analytic	0.84	1.35	2.30	1.55	2.56	0.54	3.10
Nonroutine interactive	1.47	2.36	4.01	2.70	3.16	1.04	4.20
Routine cognitive	−1.05	−1.68	−2.86	−1.92	−3.14	−3.32	−6.46
Routine manual	−1.15	−1.86	−3.15	−2.12	−2.84	0.32	−2.52
C. 10 × predicted annual changes in college-equivalent share of employment in percentage points, due to observed task shifts (panel A)							
Nonroutine tasks	1.53	1.40	1.63	1.51	1.53	−0.36	0.83
Routine tasks	−0.20	0.66	1.17	0.50	0.97	0.27	0.87
All tasks	1.33	2.06	2.80	2.01	2.49	−0.09	2.40

Panel A: Observed extensive margin task shifts are defined as the change in economywide input of each task (in percentiles of the 1960 task distribution) estimated using DOT 1977 occupational task measures applied to Census and CPS samples for 1970 to 1998 and summarized in Table II. Intensive margin shifts are measured as change in mean of DOT occupational task input (measured in centiles of the 1977 task distribution) between 1977 and 1991 DOT revisions, using the 1980 and 1998 occupational distributions of employment from the CPS MORG samples. See Table I and Appendix 1 for definitions and examples of task variables.

The number to beat: college demand grew 3.9 log points a year

	1. 1970–1980 extensive margin	2. 1980–1990 extensive margin	3. 1990–1998 extensive margin	4. 1970–1998 extensive margin	5. 1980–1998 extensive margin	6. 1980–1998 intensive margin	7. 1980–1998 extensive + intensive
	D. 10× annual changes in college-equivalent share of employment in percentage points, predicted by impact of computerization on task input (panel B)						
	NIPA computer investment measure				CPS computer use measure		
Nonroutine tasks	0.40	0.65	1.10	0.69	1.41	0.40	1.81
Routine tasks	0.29	0.48	0.81	0.51	0.80	1.04	1.84
All tasks	0.70	1.12	1.91	1.19	2.21	1.44	3.65
	E. Estimated log demand shifts for college-equivalent/noncollege-equivalent labor 1970–1998 (100 × annual log changes)						
	Using constant-elasticity of substitution model to estimate changes in college demand						
$\sigma = 0.0$	4.99	2.53	2.25	3.33			2.41
$\sigma = 1.4$	3.95	4.65	2.76	3.86			3.81
$\sigma = 2.0$	3.50	5.56	2.98	4.09			4.41
	Using task model to predict changes in college demand						
Total task Δ (panel C)	1.23	1.29	1.43	1.31	1.56	−0.06	1.51
Predicted by computer-ization (panel D)	0.64	0.70	0.98	0.76	1.39	0.91	2.29

Observed task change explains about a third of it: $1.31/3.86 \approx 34\%$

	1. 1970–1980 extensive margin	2. 1980–1990 extensive margin	3. 1990–1998 extensive margin	4. 1970–1998 extensive margin	5. 1980–1998 extensive margin	6. 1980–1998 intensive margin	7. 1980–1998 extensive + intensive
	D. 10× annual changes in college-equivalent share of employment in percentage points, predicted by impact of computerization on task input (panel B)						
	NIPA computer investment measure				CPS computer use measure		
Nonroutine tasks	0.40	0.65	1.10	0.69	1.41	0.40	1.81
Routine tasks	0.29	0.48	0.81	0.51	0.80	1.04	1.84
All tasks	0.70	1.12	1.91	1.19	2.21	1.44	3.65
	E. Estimated log demand shifts for college-equivalent/noncollege-equivalent labor 1970–1998 (100 × annual log changes)						
	Using constant-elasticity of substitution model to estimate changes in college demand						
$\sigma = 0.0$	4.99	2.53	2.25	3.33			2.41
$\sigma = 1.4$	3.95	4.65	2.76	3.86			3.81
$\sigma = 2.0$	3.50	5.56	2.98	4.09			4.41
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Total task Δ (panel C)	1.23	1.29	1.43	1.31	1.56	−0.06	1.51
Predicted by computer-ization (panel D)	0.64	0.70	0.98	0.76	1.39	0.91	2.29

Add within-occupation change and computers alone explain 2.29/3.81 ≈ 60%

	1. 1970–1980 extensive margin	2. 1980–1990 extensive margin	3. 1990–1998 extensive margin	4. 1970–1998 extensive margin	5. 1980–1998 extensive margin	6. 1980–1998 intensive margin	7. 1980–1998 extensive + intensive
	D. 10× annual changes in college-equivalent share of employment in percentage points, predicted by impact of computerization on task input (panel B)						
	NIPA computer investment measure				CPS computer use measure		
Nonroutine tasks	0.40	0.65	1.10	0.69	1.41	0.40	1.81
Routine tasks	0.29	0.48	0.81	0.51	0.80	1.04	1.84
All tasks	0.70	1.12	1.91	1.19	2.21	1.44	3.65
	E. Estimated log demand shifts for college-equivalent/noncollege-equivalent labor 1970–1998 (100 × annual log changes)						
	Using constant-elasticity of substitution model to estimate changes in college demand						
$\sigma = 0.0$	4.99	2.53	2.25	3.33			2.41
$\sigma = 1.4$	3.95	4.65	2.76	3.86			3.81
$\sigma = 2.0$	3.50	5.56	2.98	4.09			4.41
	Using task model to predict changes in college demand						
Total task Δ (panel C)	1.23	1.29	1.43	1.31	1.56	−0.06	1.51
Predicted by computer-ization (panel D)	0.64	0.70	0.98	0.76	1.39	0.91	2.29

Caveat: observed within-occupation task change contributed nothing (−0.06)

	1. 1970–1980 extensive margin	2. 1980–1990 extensive margin	3. 1990–1998 extensive margin	4. 1970–1998 extensive margin	5. 1980–1998 extensive margin	6. 1980–1998 intensive margin	7. 1980–1998 extensive + intensive
	D. 10× annual changes in college-equivalent share of employment in percentage points, predicted by impact of computerization on task input (panel B)						
	NIPA computer investment measure				CPS computer use measure		
Nonroutine tasks	0.40	0.65	1.10	0.69	1.41	0.40	1.81
Routine tasks	0.29	0.48	0.81	0.51	0.80	1.04	1.84
All tasks	0.70	1.12	1.91	1.19	2.21	1.44	3.65
	E. Estimated log demand shifts for college-equivalent/noncollege-equivalent labor 1970–1998 (100 × annual log changes)						
	Using constant-elasticity of substitution model to estimate changes in college demand						
$\sigma = 0.0$	4.99	2.53	2.25	3.33			2.41
$\sigma = 1.4$	3.95	4.65	2.76	3.86			3.81
$\sigma = 2.0$	3.50	5.56	2.98	4.09			4.41
	Using task model to predict changes in college demand						
Total task Δ (panel C)	1.23	1.29	1.43	1.31	1.56	−0.06	1.51
Predicted by computer-ization (panel D)	0.64	0.70	0.98	0.76	1.39	0.91	2.29

Bottom line: how much of KM's demand shift is tasks?

- KM-style benchmark, 1970–98, $\sigma = 1.4$: college demand rose 3.9 log points per year. That is the black box
- Observed task shifts (extensive margin) deliver 1.31 $\rightarrow$ about a third of it (39% against the more comparable $\sigma = 0$ benchmark of 3.33)
- For 1980–98, adding within-occupation task change and using only the computer-predicted part: 2.29 against 3.81 $\rightarrow$ 60% (95% against $\sigma = 0$)

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- Two honest caveats: (i) computer-predicted change (2.29) exceeds total observed change (1.51), since non-computer forces pushed the other way within occupations; (ii) fixed $\widehat{\pi}_k$ means all of these are lower bounds
- Verdict: tasks are not the whole story, but a single measurable mechanism accounts for a third to a half of something KM could only name

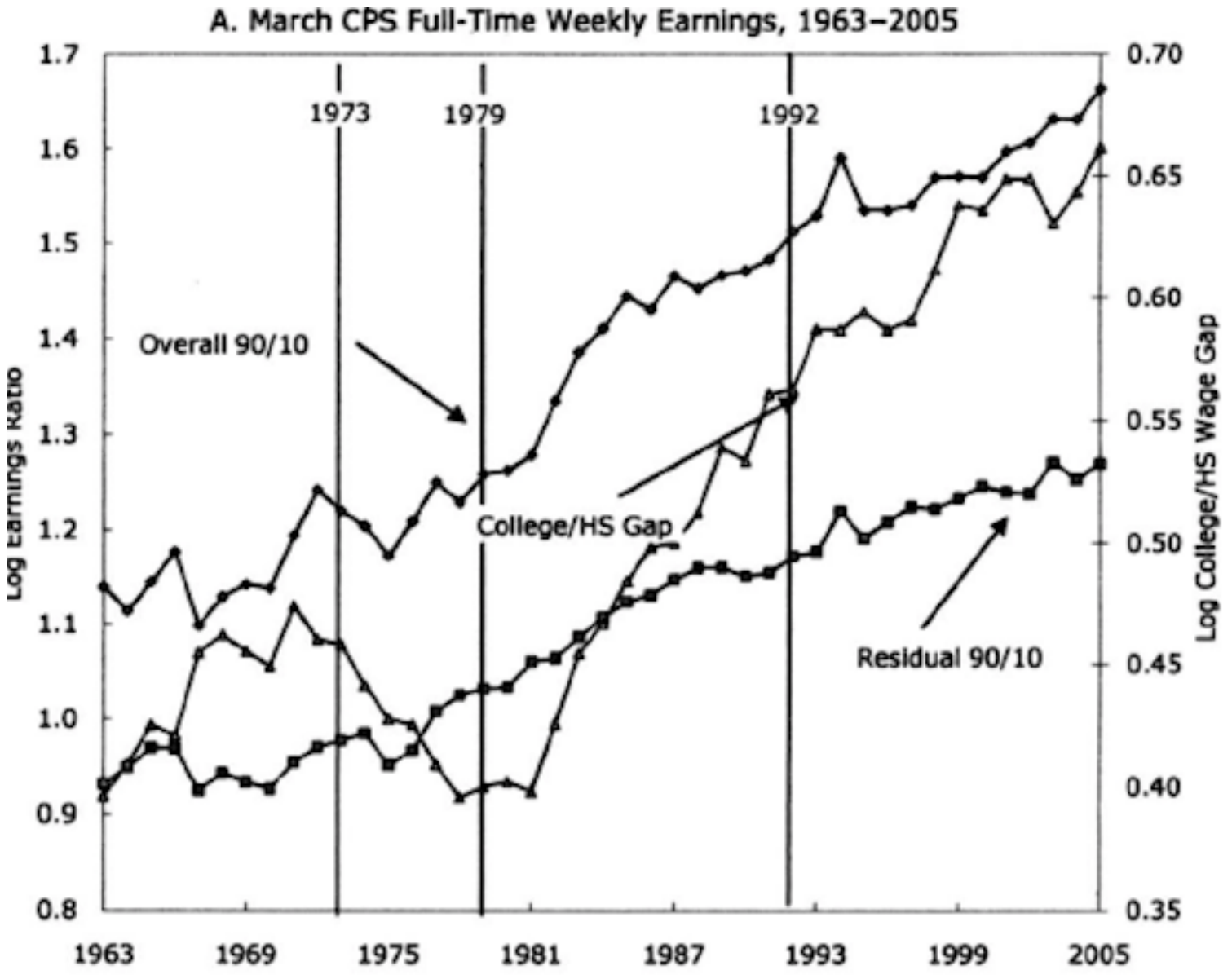
Summary

- Influential reboot of SBTC narrative ($\approx$10000 GS cites)
- Task framework clarifies how computerization influences wages
- Using DOT to measure tasks provides additional empirical content about mechanisms underlying demand side
- Still very indirect evidence of technical change
  - Moving the goal posts: now we are explaining industry or occupation level moments.
  - How important are occupations for overall wage inequality? (Firpo, Fortin, Lemieux, 2009)
- See also Acemoglu and Autor HOLE (2011) for a review of task-based approach in economics.

Autor Katz Kearney (2008)

- Actual title: "Revising the Revisionists"
- Two claims of the Revisionists are under scrutiny:
  1. Growth of inequality was episodic rather than a secular phenomenom.
  2. Such episodic rise is largely explained by non-market forces (aka institutions) and labor force composition changes.
- Data: CPS (March + ORG). Series updated up to 2005.
- Potential inconsistencies with ORG CPS due to the major restructuring of the earnings questions that occurred in 1994.

Where is the agreement? Where is the disagreement?

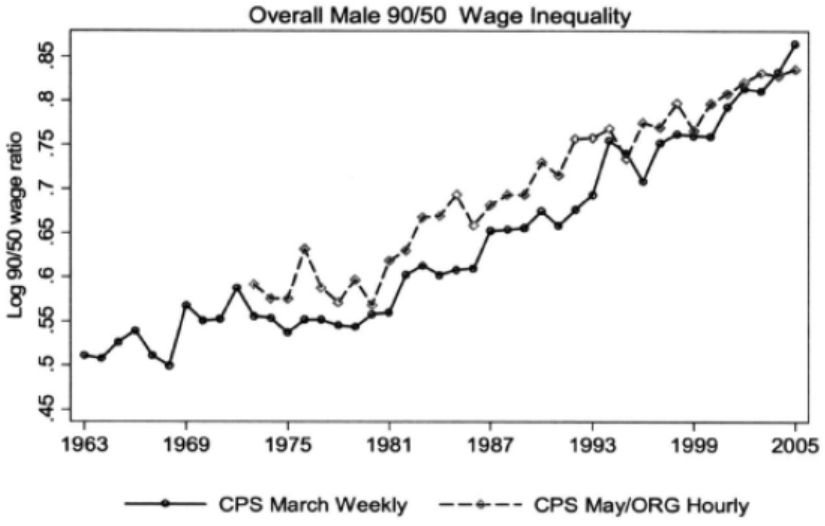


Revisiting trends

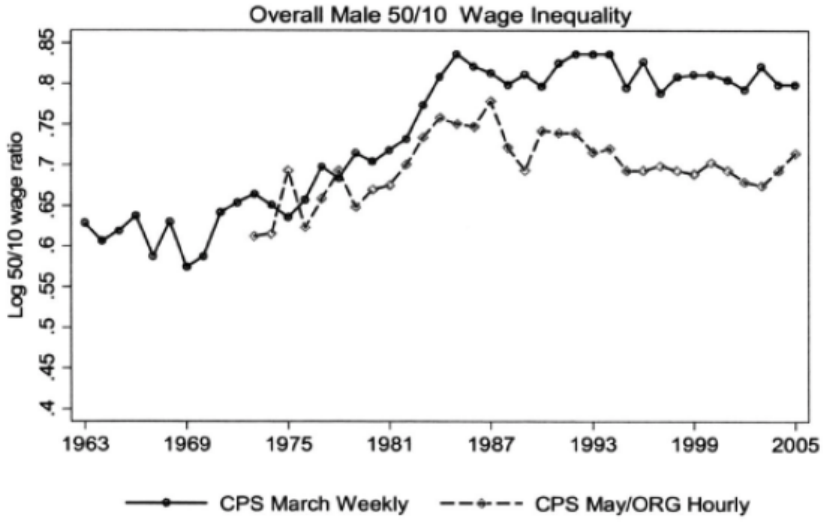
- 90/10 log wage differential, which is the difference in real log wages between the 90th and the 10th percentile of the wage distribution.
- 90/50 - 50/10 log wage differential which measure upper tail and lower tail inequality
- College-HS log wage gap (between group inequality)
- 90/10, 90/50, 50/10 residual log wage gaps which are constructed after partialling out education, experience and gender.
- Changes in residual inequality measures changes in inequality that are not explained by changes in the composition of the workforce as measured by these variables.

Episodic vs. Non Episodic Inequality

FIGURE 3.—90/50 AND 50/10 WEEKLY WAGE INEQUALITY IN MARCH (FULL-TIME WORKERS) AND HOURLY WAGE INEQUALITY IN MAY/ORG (ALL WORKERS) CPS SERIES, 1963–2005





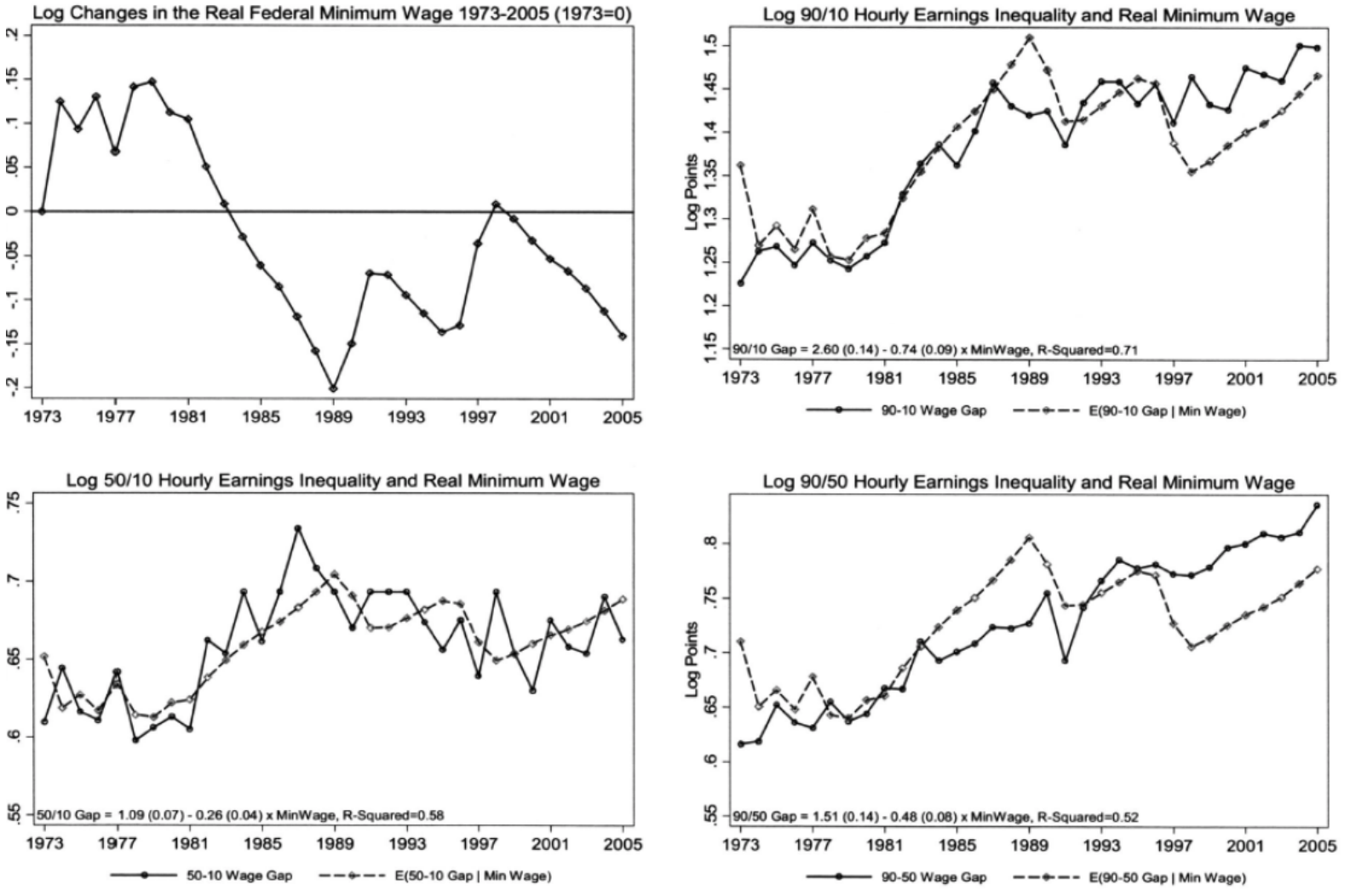




See notes to figure 2 for details on samples and data processing.

Min Wage plausible for lower tail inequality but what about upper tail?

FIGURE 7.—LOG REAL FEDERAL MINIMUM WAGE AND LOG 90/10, 90/50, AND 50/10 HOURLY WAGE DIFFERENTIALS, 1973–2005 (CPS MAY/ORG).



minal minimum wages are deflated to real log values using the PCE deflator. In the first panel, the real log minimum-wage measure is normalized to zero in 1973. Subsequent panels depict the observed wage gap (90/10, 90/50, and 50/10) for all hourly workers the CPS May/ORG samples in each year plotted alongside the predicted values from separate OLS regressions of the relevant wage gap on a constant and the contemporaneous real log minimum wage.

$\sigma = 1.4$ even after controlling for minimum wage

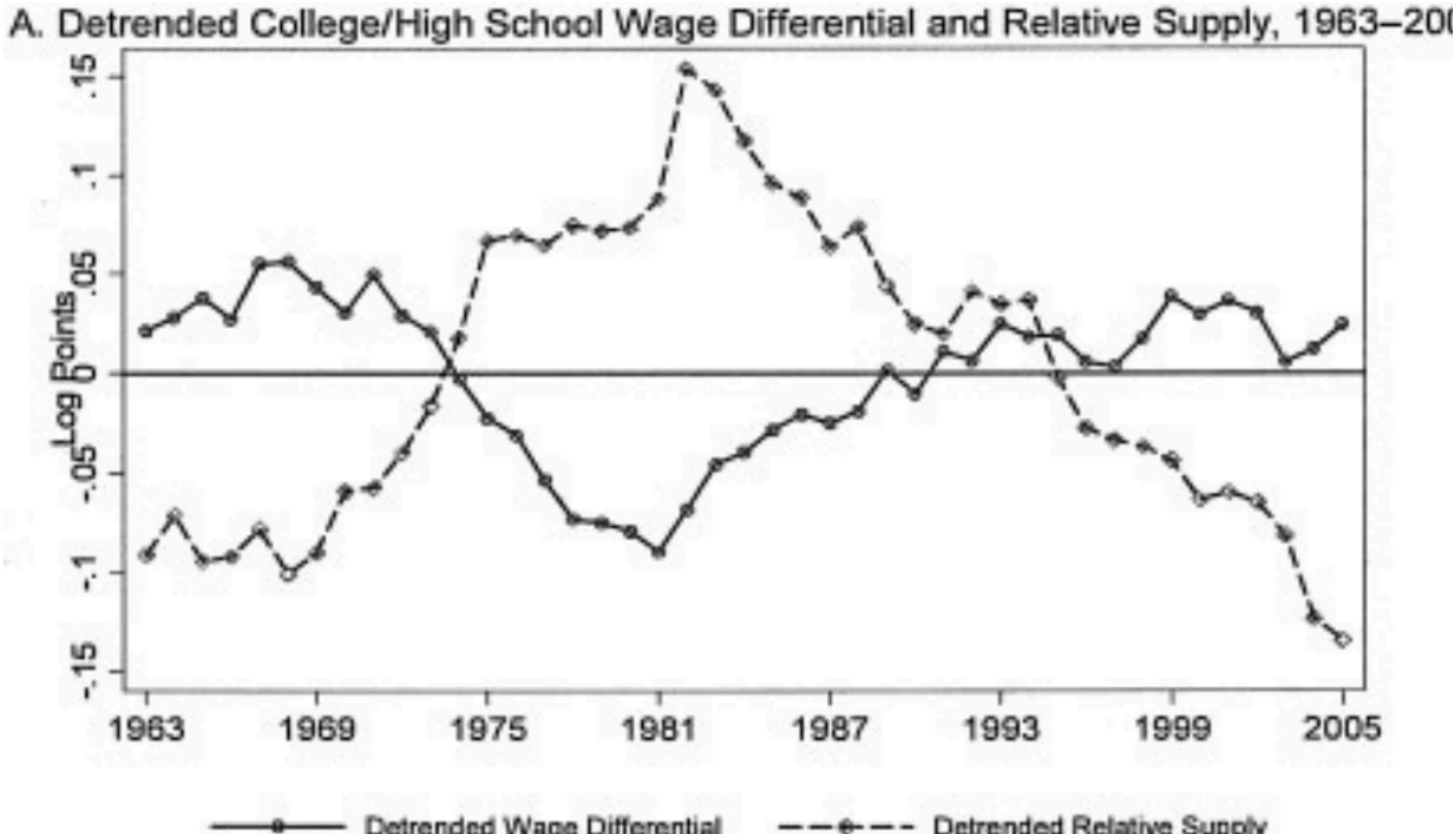
TABLE 2.—REGRESSION MODELS FOR THE COLLEGE/HIGH SCHOOL LOG WAGE GAP, 1963–2005

	(1) 1963–1987	(2)	(3)	(4)	(5) 1963–2005	(6)	(7)	(8)
CLG/HS relative supply	−0.636	−0.411	−0.619	−0.599	−0.609	−0.728	−0.403	
	(0.130)	(0.046)	(0.066)	(0.112)	(0.102)	(0.155)	(0.067)	
Log real minimum wage						−0.049	−0.117	−0.144
						(0.051)	(0.047)	(0.065)
Male prime-age unemp. rate						0.004	−0.001	−0.018
						(0.004)	(0.004)	(0.003)
Time	0.026	0.018	0.026	0.028	0.021	0.028	0.017	0.006
	(0.005)	(0.001)	(0.002)	(0.006)	(0.006)	(0.007)	(0.002)	(0.001)
Time^2/100				−0.011	0.030	0.017		
				(0.006)	(0.015)	(0.017)		
Time^3/1000					−0.006	−0.005		
					(0.002)	(0.002)		
Time × post-1992			−0.008					
			(0.002)					
Constant	−0.159	0.043	−0.146	−0.143	−0.124	−0.160	0.266	0.689
	(0.119)	(0.037)	(0.057)	(0.108)	(0.098)	(0.191)	(0.112)	(0.120)
Observations	25	43	43	43	43	43	43	43
R-squared	0.563	0.934	0.953	0.940	0.952	0.955	0.944	0.891

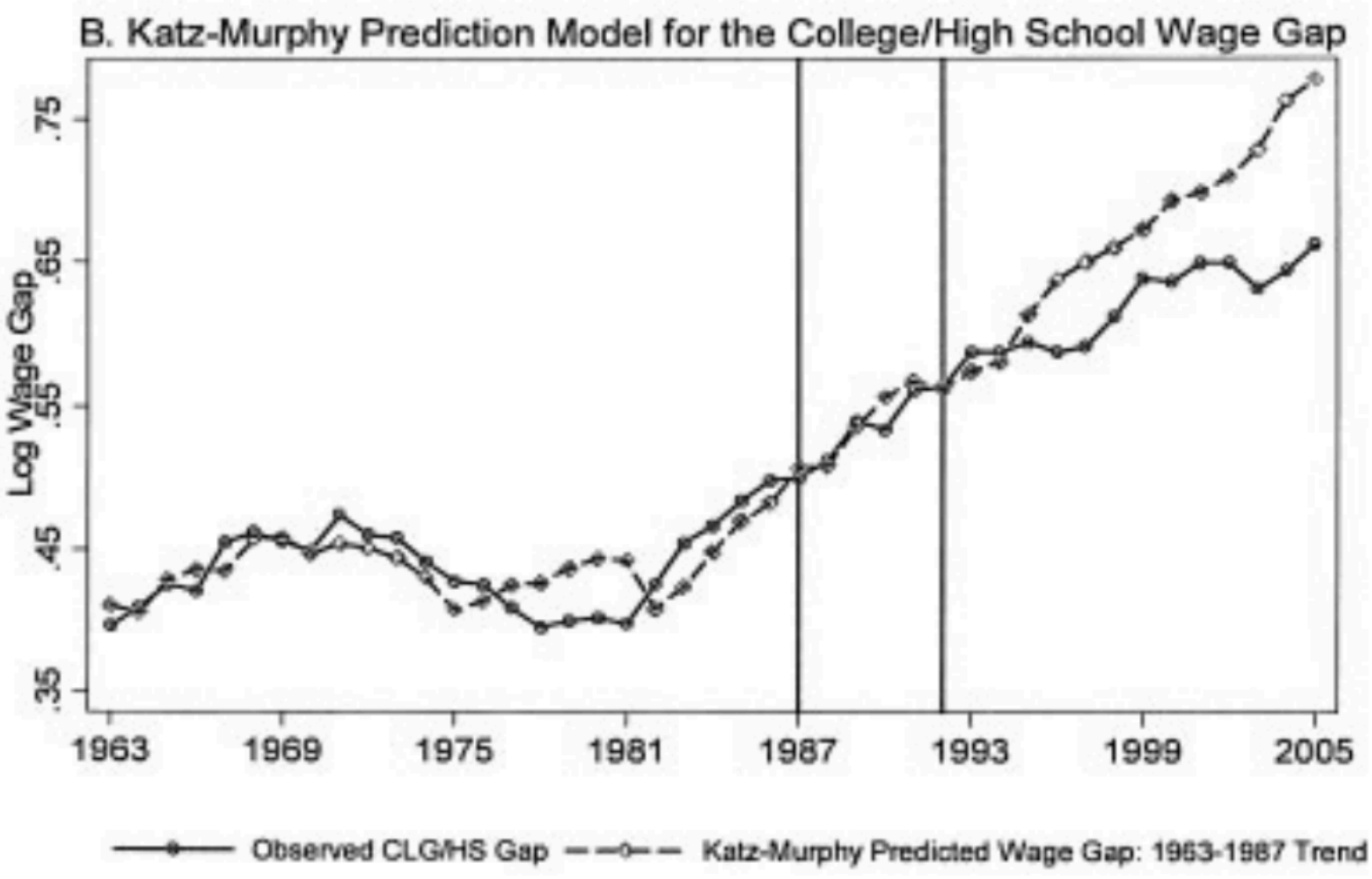
Standard errors in parentheses. Each column presents an OLS regression of the fixed-weighted college/high school wage differential on the indicated variables. The U.S. federal minimum wage is deflated by the personal consumption expenditure deflator. Source for labor supply and earnings measures is the March CPS, earnings years 1963–2005.

Extending KM up to 2005....

FIGURE 4.—COLLEGE/HIGH SCHOOL RELATIVE SUPPLY AND WAGE DIFFERENTIAL, 1963–2005 (MARCH CPS)



Extending KM up to 2005....



AKK Facts

- Inequality grew at a lower pace in the 1990s, early 2000s than in the 1980s.
- Lower tail inequality largely explained by the minimum wage.
- But grow in inequality is not an episodic event! Upper tail inequality continued to increase in recent years while inequality in lower half stabilized somewhat.
- Common theme: polarization. After a monotone surge in inequality from 1979-1987, upper tail inequality continued to grew while lower tail inequality flatted out or even was reduced.

But what explains polarization in inequality?

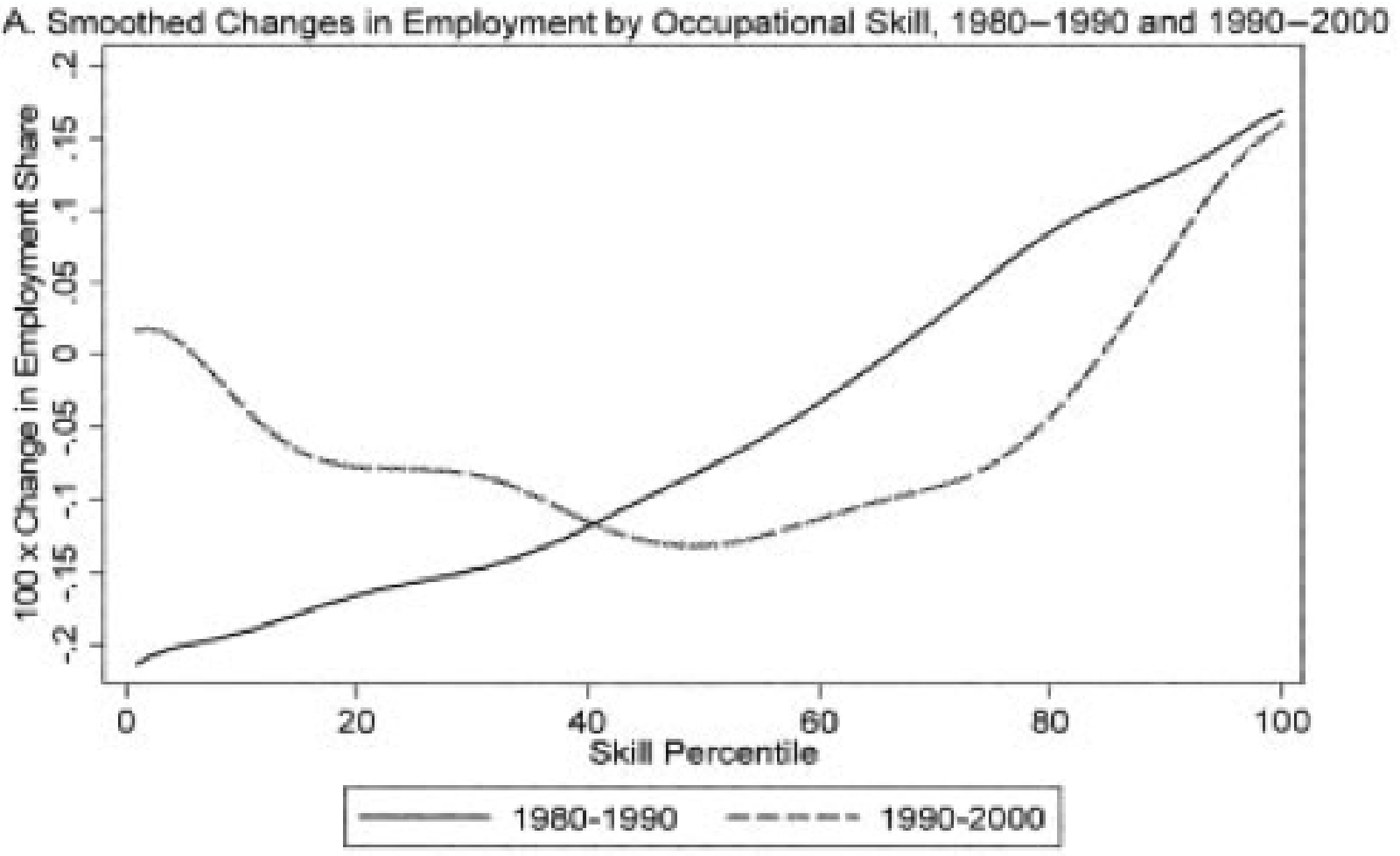
- Not due to mechanical shifts in the composition of the workforce.
- Not due to minimum wages.
- Canonical supply-demand models also do not fully capture what is going on post 1992.

But what explains polarization in inequality?

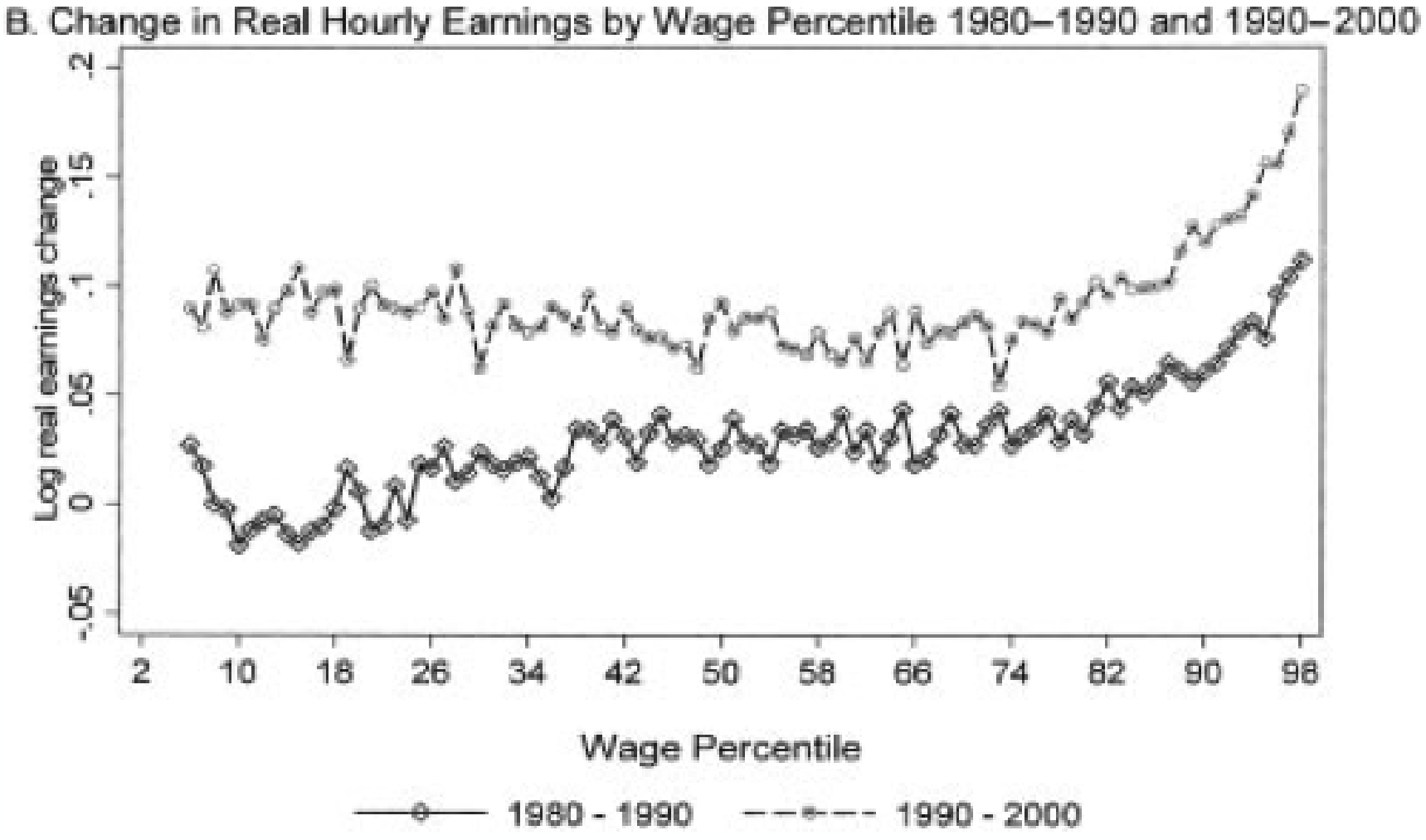
- Not due to mechanical shifts in the composition of the workforce.
- Not due to minimum wages.
- Canonical supply-demand models also do not fully capture what is going on post 1992.
- New concept: Tasked based production function (Autor-Levy-Murnane, 2003)
  - Computerization has non-monotone effects on demand for skills throughout the earnings distribution.
  - Sharp raise in the demand for abstract task (complementary).
  - Reducing demand for clerical, mechanical routine tasks done by individuals in the middle of the earnings distribution (substitution).
  - But no impact on manual tasks! (still need janitors)

Census data + Dictionary of Occupational Titles

FIGURE 11.—(A) CHANGE IN OCCUPATION'S EMPLOYMENT SHARES, 1980–1990 AND 1990–2000, BY OCCUPATIONAL SKILL PERCENTILE IN 1980 (B) CHANGES IN REAL HOURLY EARNINGS BY WAGE PERCENTILE, 1980–1990 AND 1990–2000 (CENSUS 1980, 1990, AND 2000, ALL HOURLY WORKERS)



Smoking gun?


B. Change in Real Hourly Earnings by Wage Percentile 1980–1990 and 1990–2000
Log real earnings change
-.05
0
.05
.1
.15
.2
2
10
18
26
34
42
50
58
66
74
82
90
98
Wage Percentile
1980 - 1990
1990 - 2000

Summarizing the saga...

- (Almost) Unanimous consensus upon the following facts
  - Upper and lower tail inequality exhibit different time series properties.
  - Relative supply shifts matters.
  - Minimum wage important for lower tail inequality
  - Occupational task content models does a good job in predicting polarization in inequality in the 1990s.
  - Observables far from explaining all changes to inequality.

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  - Relative supply shifts matters.
  - Minimum wage important for lower tail inequality
  - Occupational task content models does a good job in predicting polarization in inequality in the 1990s.
  - Observables far from explaining all changes to inequality.
- Immense literature that generated several "new" frontiers
  - Minimum Wage.
  - Trade (China Shocks $\rightarrow$Autor, Dorn and Hanson)
  - Firms as "institutions".
  - Impacts of Robots/AI.

Foundational Questions

- US-centric literature (Dustmann, Ludsteck, Schonberg, 2009).
- Largely based on survey data. Key revolution of the 2010s is to bring tax and or administrative records to this literature.
- Aggregate production function / cost functions?
  - How to incorporate realistic heterogeneity?
  - What are the limitations of Nested CES models?
- Always assumed that labor markets are competitive.
  - Maybe different degrees of imperfect competition over-time have role in explaining wage inequality?
  - How to incorporate frictions in S-D-I models? See Daniel Haanwinckel JMP.

Applying AI to Rebuild Middle Class Jobs
David Autor
NBER Working Paper No. 32140
February 2024
JEL No. J01,J2,N30,O14

ABSTRACT

While the utopian vision of the current Information Age was that computerization would flatten economic hierarchies by democratizing information, the opposite has occurred. Information, it turns out, is merely an input into a more consequential economic function, decision-making, which is the province of elite experts. The unique opportunity that AI offers to the labor market is to extend the relevance, reach, and value of human expertise. Because of AI's capacity to weave information and rules with acquired experience to support decision-making, it can be applied to enable a larger set of workers possessing complementary knowledge to perform some of the higher-stakes decision-making tasks that are currently arrogated to elite experts, e.g., medical care to doctors, document production to lawyers, software coding to computer engineers, and undergraduate education to professors. My thesis is not a forecast but an argument about what is possible: AI, if used well, can assist with restoring the middle-skill, middle-class heart of the US labor market that has been hollowed out by automation and globalization.

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Project Syndicate

Don’t Believe the AI Hype

May 21, 2024 |**DARON ACEMOGLU**

BOSTON – According to tech leaders and many pundits and academics, artificial intelligence is poised to transform the world as we know it through unprecedented productivity gains. While some believe that machines soon will do everything humans can do, ushering in a new age of boundless prosperity, other predictions are at least more grounded. For example, Goldman Sachs predicts that generative AI will boost global GDP by 7% over the next decade, and the McKinsey Global Institute anticipates that the annual GDP growth rate could increase by 3-4 percentage points between now and 2040. For its part, *The Economist* expects that AI will create a blue-collar bonanza.
Is this realistic? As I note in a recent paper, the outlook is far more uncertain than most forecasts and guesstimates suggest. Still, while it is basically impossible to predict with any confidence what AI will do in 20 or 30 years, one can say something about the next decade, because most of these near-term economic effects must involve existing technologies and improvements to them.

It is reasonable to suppose that AI’s biggest impact will come from automating some tasks and making some workers in some occupations more productive. Economic theory provides some guidance for assessing these aggregate effects. According to Hulten’s theorem (named for economist Charles Hulten), aggregate “total factor productivity” (TFP) effects are simply the product of the share of tasks that are automated multiplied by the average cost savings.

While average cost savings are difficult to estimate and will vary by activity, there have already been some careful studies of AI’s effects on certain tasks. For example, Shakked Noy and Whitney Zhang have examined the impact of ChatGPT on simple writing tasks (such as summarizing documents or writing routine grant proposals or marketing material), while Erik Brynjolfsson, Danielle Li, and Lindsey Raymond have assessed the use of AI assistants in customer service. Taken together, this research suggests that currently available generative-AI tools yield average labor-cost savings of 27% and overall cost savings of 14.4%.

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What about the share of tasks that will be affected by AI and related technologies? Using numbers from recent studies, I estimate this to be around 4.6%, implying that AI will increase TFP by only

New Frontiers

0.66% over ten years, or by 0.06% annually. Of course, since AI will also drive an investment boom, the increase in GDP growth could be a little larger, perhaps in the 1-1.5% range.

These figures are much smaller than the ones from Goldman Sachs and McKinsey. If you want to get those bigger numbers, you either must boost the productivity gains at the micro level or assume that many more tasks in the economy will be affected. But neither scenario seems plausible. Labor-cost savings far above 27% not only fall out of the range offered by existing studies; they also do not align with the observed effects of other, even more promising technologies. For example, industrial robots have transformed some manufacturing sectors, and they appear to have reduced labor costs by about 30%.
Similarly, we are unlikely to see far more than 4.6% of tasks being taken over, because AI is nowhere close to being able to perform most manual or social tasks (including seemingly simple functions with some social aspects, like accounting). As of 2019, a survey of essentially all US businesses found that only about 1.5% of them had any AI investments. Even if such investments have picked up over the past year and a half, we have a long, long way to go before AI becomes widespread.